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Some approximation results for superhedging prices via neural networks

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Chapter 1

Introduction

Contingent claims are derivative financial instruments whose value depends on a group of underlying financial assets and their price fluctuation. Thereby, one distinguishes between American and European contingent claims; while the buyer of an American claim can exercise the claim at any time up to the expiration of the contract, the payoff of a European contingent claim is due at a fixed maturity date. In the latter case, the goal of the seller is to hedge against the claim, that is, to find an investment strategy that can offset the risk of a potential loss from selling the claim. If the market is incomplete, it is not possible to replicate the exact strike price of every contingent claim. Superhedging provides an alternative, with the idea of finding a strategy with minimal initial investment that covers any obligation resulting from the sale of the claim. However, the superhedging price often results too high from both a theoretical and a practical point of view, thus reducing the probability of making a profit. The method of quantile hedging aims to solve this problem, where the seller of the claim can either minimize the initial cost by accepting a shortfall probability or maximize the probability of superhedging for a given initial capital.

In general incomplete markets, both the superhedging and quantile hedging prices can result difficult to compute explicitly. Hence, the purpose of this thesis is on the one hand to introduce part of the theory available on both of the aforementioned methods, and on the other hand to approximate the superhedging and quantile hedging prices via neural networks. Feedforward neural networks are models loosely inspired by the human brain, which represent an essential machine-learning tool. Due to their excellent approximation properties, neural networks provide a framework for the approximation of a large class of functions.

We consider a financial market model with finitely many trading dates and finitely many assets. Price fluctuations and trading strategies are represented by stochastic processes on a filtered probability space $(\Omega, \mathcal{F}, \mathbb{F}, \mathbb{P})$. Moreover, we assume that our market is arbitrage-free, meaning that there exist no investment opportunities without any downside risk. By the first fundamental theorem of asset pricing, this assumption is equivalent to the existence of an equivalent martingale measure, that is, a probability measure equivalent to $\mathbb{P}$ under which the price process is a martingale. A European contingent claim is

represented by a nonnegative random variable H . If the market is complete, H admits a unique arbitrage-free price. In the case of an incomplete market, we show that there exists a trading strategy whose final value is greater than or equal to the (random) payoff of the claim H at the expiration of the contract. Furthermore, we show that the initial investment needed for the purchase of this strategy, called the superhedging price, is minimal under all superhedging strategies. One important consideration is that in general, the superhedging price is strictly higher than any arbitrage-free price for the claim H , therefore eliminating the opportunity of making a profit from the seller. We present quantile hedging as a solution to this problem. For a fixed $\alpha \in (0, 1)$, we consider the problem of finding a strategy with minimal initial investment which covers the cost of selling H with probability greater than α . We prove that for α tending to 1, this reduces to the superhedging problem. Moreover, we prove the existence of a solution to an extended version of the α -quantile hedging problem, via a specific kind of randomized tests, called success ratios.

For the approximation results contained in this work, we introduce feedforward neural networks and recall their universal approximation properties, in particular on spaces of random variables. To approximate the superhedging and α -quantile hedging prices, we assume that the filtration $\mathbb{F}$ is generated by some multi-dimensional stochastic process. Under this assumption, we can approximate adapted and predictable processes arbitrarily well by neural networks. Hence, we construct the approximate superhedging and quantile hedging problems, where the trading strategies are represented by neural networks. Finally, we show that under a suitable assumption, the respective solutions of these problems approximate the original superhedging and α -quantile hedging prices.

Our work is structured as follows. In Chapter 2, we begin with some preliminaries from probability theory and present the multi-period financial market model in discrete time. Chapter 3 introduces the concept of superhedging for a European contingent claim H . We then approach the method of quantile hedging in Chapter 4 first via success sets, and then passing to success ratios. Finally, Chapter 5 contains the neural network approximations of the superhedging price process of the claim H .

Chapter 2

Preliminaries

This chapter aims to provide the reader with the necessary knowledge of probability theory and mathematical finance to understand the theory later presented in this work. Section 2.1 recalls a few important concepts about stochastic processes in discrete time. In Section 2.2, we present the finite-discrete-time financial market model that is used throughout our work, where the asset prices are modeled by a nonnegative stochastic process on a filtered probability space. Finally, we introduce the reader to arbitrage theory.

2.1 Probability theory

In this section, we recall important notions from probability theory and introduce some results for later purposes. We state the monotone and dominated convergence theorems, define the essential supremum and martingales, and present a useful convergence result for bounded sequences of random variables.

Throughout this section, we work on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. For a real-valued random variable

$$X : \Omega \rightarrow \mathbb{R},$$

$\mathbb{E}[X]$ denotes the expectation of X under $\mathbb{P}$. The expectation of a random vector $X = (X_1, \dots, X_d) : \Omega \rightarrow \mathbb{R}^d$ is then given by $\mathbb{E}[X] = (\mathbb{E}[X_1], \dots, \mathbb{E}[X_d])$. We recall that a *filtration* on $(\Omega, \mathcal{F})$ is a family of σ -fields $\mathbb{F} = (\mathcal{F}_t)_{t=0, \dots, T}$, $\mathcal{F}_t \subseteq \mathcal{F}$, such that $\mathcal{F}_0 \subseteq \mathcal{F}_1 \subseteq \dots \subseteq \mathcal{F}_T$. We often think of $\mathcal{F}_t$ as the information available at time t .

Definition 2.1. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be endowed with a filtration $\mathbb{F} = (\mathcal{F}_t)_{t=0, \dots, T}$.

- (i) A stochastic process $X = (X_t)_{t=0, \dots, T}$ is called $(\mathbb{F})$ -*adapted* if X_t is $\mathcal{F}_t$ -measurable for $t = 0, \dots, T$.
- (ii) A stochastic process $Y = (Y_t)_{t=1, \dots, T}$ is called $(\mathbb{F})$ -*predictable* if Y_t is $\mathcal{F}_{t-1}$ -measurable for $t = 1, \dots, T$.

We denote by $\mathcal{L}^0 = \mathcal{L}^0(\Omega, \mathcal{F}, \mathbb{P})$ the set of random variables on Ω . Then $L^0 = L^0(\Omega, \mathcal{F}, \mathbb{P}) := \mathcal{L}^0(\Omega, \mathcal{F}, \mathbb{P}) / \sim$ is the set of equivalence classes of random variables in $\mathcal{L}^0$ with respect to the equivalence relation

$$X \sim Y \quad \Leftrightarrow \quad X = Y \text{ } \mathbb{P}\text{-almost everywhere.} \quad (2.1)$$

Similarly, for $p \in [1, \infty)$ we write

$$\mathcal{L}^p(\Omega, \mathcal{F}, \mathbb{P}) := \{X \in \mathcal{L}^0 : \mathbb{E}[|X|^p] < \infty\},$$

and we denote the set of bounded random variables by

$$\mathcal{L}^\infty := \{X \in \mathcal{L}^0 : \exists K \in \mathbb{R} \text{ s.t. } |X| \leq K\}.$$

Then we let

$$L^p(\Omega, \mathcal{F}, \mathbb{P}) := \mathcal{L}^p(\Omega, \mathcal{F}, \mathbb{P})/\sim \quad \text{and} \quad L^\infty(\Omega, \mathcal{F}, \mathbb{P}) := \mathcal{L}^\infty(\Omega, \mathcal{F}, \mathbb{P})/\sim,$$

where $\sim$ is defined as in (2.1). We say that a random variable X is $(\mathbb{P})$ -integrable if $X \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$. Finally, for a random variable taking values in some set $U \subseteq \mathbb{R}^d$, we sometimes write $X \in L^p(\Omega, \mathcal{F}, \mathbb{P}; U)$, for $p \in [1, \infty]$.

Next, we recall three important convergence results for the expectation of sequences of random variables. For the proofs, we refer the reader to Durrett [4, Theorems 1.5.5, 1.5.7 and 1.5.8].

Theorem 2.2 (Fatou's lemma). *Let $(X_n)_{n \in \mathbb{N}}$ be a nonnegative sequence of random variables. Then*

$$\liminf_{n \rightarrow \infty} \mathbb{E}[X_n] \geq \mathbb{E}[\liminf_{n \rightarrow \infty} X_n].$$

Theorem 2.3 (Monotone convergence theorem). *Let $(X_n)_{n \in \mathbb{N}}$ be a nonnegative increasing sequence of random variables, and let $X \geq 0$ be a random variable such that $\lim_{n \rightarrow \infty} X_n = X$. Then*

$$\lim_{n \rightarrow \infty} \mathbb{E}[X_n] = \mathbb{E}[X].$$

Theorem 2.4 (Dominated convergence theorem). *Let $(X_n)_{n \in \mathbb{N}}$ be a sequence of random variables such that $\lim_{n \rightarrow \infty} X_n = X$ $\mathbb{P}$ -a.s. for some $X \in L^1(\Omega, \mathcal{F}, \mathbb{P})$ with $X \geq 0$. If $|X_n| \leq X$ $\mathbb{P}$ -a.s. for each $n \in \mathbb{N}$, then*

$$\lim_{n \rightarrow \infty} \mathbb{E}[X_n] = \mathbb{E}[X].$$

For a random variable $X \in L^1(\Omega, \mathcal{F}, \mathbb{P})$ and a sub- σ -algebra $\mathcal{G} \subseteq \mathcal{F}$, we denote the conditional expectation of X given $\mathcal{G}$ by $\mathbb{E}[X|\mathcal{G}] \in L^1(\Omega, \mathcal{G}, \mathbb{P})$. Note that Theorems 2.2–2.4 above can be extended to conditional expectations. We recall the following properties of the conditional expectation.

- (i) Let $X \in L^1(\Omega, \mathcal{F}, \mathbb{P})$ and let $\mathcal{G} \subseteq \mathcal{F}$ be a trivial sub- σ -algebra, i.e., $\mathbb{P}[A] \in \{0, 1\}$ for all $A \in \mathcal{G}$. Then

$$\mathbb{E}[X|\mathcal{G}] = \mathbb{E}[X].$$

(ii) Let $\mathcal{G} \subseteq \mathcal{F}$ be a sub- σ -algebra and let $X \in L^1(\Omega, \mathcal{G}, \mathbb{P})$. Then

$$\mathbb{E}[X|\mathcal{G}] = X \quad \mathbb{P}\text{-a.s.}$$

(iii) Let $\mathcal{A} \subseteq \mathcal{G} \subseteq \mathcal{F}$ be sub- σ -algebras and let $X \in L^1(\Omega, \mathcal{F}, \mathbb{P})$. Then

$$\mathbb{E}[\mathbb{E}[X|\mathcal{G}]|\mathcal{A}] = \mathbb{E}[X|\mathcal{A}] \quad \mathbb{P}\text{-a.s.}$$

This is often called *tower property* for conditional expectations.

(iv) Let $Z \in L^\infty(\Omega, \mathcal{F}, \mathbb{P})$ be bounded and $X \in L^1(\Omega, \mathcal{F}, \mathbb{P})$. Then

$$\mathbb{E}[ZX|\mathcal{F}] = Z\mathbb{E}[X|\mathcal{F}] \quad \mathbb{P}\text{-a.s.}$$

Definition 2.5. A real-valued stochastic process $M = (M_t)_{t=0,\dots,T}$ on a filtered probability space $(\Omega, \mathcal{F}, \mathbb{F}, \mathbb{P})$ is called a *martingale* if

- (a) $M_t \in L^1(\Omega, \mathcal{F}_t, \mathbb{P})$ for $t = 0, \dots, T$,
- (b) $\mathbb{E}[M_t|\mathcal{F}_{t-1}] = M_{t-1}$ $\mathbb{P}$ -a.s., for $t = 1, \dots, T$.

M is called a *supermartingale* if it satisfies (a) and

- (b') $\mathbb{E}[M_t|\mathcal{F}_{t-1}] \leq M_{t-1}$ $\mathbb{P}$ -a.s., for $t = 1, \dots, T$.

Finally, M is called a *submartingale* if it satisfies (a) and

- (b'') $\mathbb{E}[M_t|\mathcal{F}_{t-1}] \geq M_{t-1}$ $\mathbb{P}$ -a.s., for $t = 1, \dots, T$.

Remark 2.6. We note that for any martingale $(M_t)_{t=0,\dots,T}$, we have

$$\mathbb{E}[M_t|\mathcal{F}_s] = M_s \quad \mathbb{P}\text{-a.s., for } 0 \leq s \leq t \leq T. \quad (2.2)$$

Indeed, since $\mathcal{F}_s \subseteq \mathcal{F}_t$ for $s \leq t$, using (a),(b) and the tower property, for any $0 \leq s < t \leq T$ we get

$$\mathbb{E}[M_t|\mathcal{F}_s] = \mathbb{E}[\mathbb{E}[M_t|\mathcal{F}_{t-1}]|\mathcal{F}_s] = \mathbb{E}[M_{t-1}|\mathcal{F}_s] = \dots = \mathbb{E}[M_s|\mathcal{F}_s] = M_s \quad \mathbb{P}\text{-a.s.,}$$

Conversely, (2.2) implies (b), so we can equivalently define a martingale by (a) and (2.2). Similarly, if M is a supermartingale or a submartingale (2.2) also holds with “ $\leq$ ” or “ $\geq$ ” instead of “ $=$ ”, respectively.

Lemma 2.8 below shows that for any bounded sequence of random variables, there exists a sequence of convex combinations converging almost surely to some random variable. This result is an important element of several proofs in Chapter 4. We refer the reader to Föllmer and Schied [9, Lemma 1.70] for its proof.

Definition 2.7. Let $(X_n)_{n \in \mathbb{N}}$ be a sequence in $L^0(\Omega, \mathcal{F}, \mathbb{P}; \mathbb{R}^d)$. The set of all finite *convex combinations* of $(X_k)_{k \geq m}$, $m \in \mathbb{N}$, is given by

$$\text{conv}(X_m, X_{m+1}, \dots) := \left\{ \sum_{k=m}^{\infty} \lambda_k X_k : \lambda_k \geq 0 \text{ for all } k \geq m, \sum_{k=m}^{\infty} \lambda_k = 1, \text{ finitely many } \lambda_k \neq 0 \right\}.$$

Lemma 2.8. Let $(X_n)_{n \in \mathbb{N}}$ be a bounded sequence in $L^0(\Omega, \mathcal{F}, \mathbb{P}; \mathbb{R}^d)$, that is, $\sup_{n \in \mathbb{N}} |X_n| < \infty$ $\mathbb{P}$ -almost surely. Then there exists a sequence of convex combinations

$$\varphi_n \in \text{conv}(X_n, X_{n+1}, \dots)$$

which converges to some $\varphi \in L^0(\Omega, \mathcal{F}, \mathbb{P}; \mathbb{R}^d)$ $\mathbb{P}$ -almost surely.

Remark 2.9. If there exists some $K \in \mathbb{R}_+$ such that the sequence $(X_n)_{n \in \mathbb{N}}$ in Lemma 2.8 additionally satisfies $|X_n| \leq K$ for each $n \in \mathbb{N}$, then by Definition 2.7 $|\varphi_n| \leq K$ for all $n \in \mathbb{N}$, and therefore also $|\varphi| \leq K$.

For finite or countably infinite families of random variables, the supremum and the infimum of that family are again random variables. However, for arbitrary families this does not always hold. The following result, Föllmer and Schied [9, Theorem A.37], guarantees the existence of a random variable that is an upper bound for such families and unique up to equality $\mathbb{P}$ -almost everywhere.

Theorem 2.10. Let Φ be any set of random variables on $(\Omega, \mathcal{F}, \mathbb{P})$.

(a) There exists a random variable φ^* which satisfies

- (i) $\varphi^* \geq \varphi$ $\mathbb{P}$ -a.s. for each $\varphi \in \Phi$.
- (ii) $\varphi^* \leq \psi$ $\mathbb{P}$ -a.s., for every random variable ψ satisfying $\psi \geq \varphi$ $\mathbb{P}$ -a.s. for each $\varphi \in \Phi$.

(b) Suppose in addition that Φ is directed upward, i.e., for $\varphi, \tilde{\varphi} \in \Phi$ there exists $\psi \in \Phi$ such that $\psi \geq \varphi \vee \tilde{\varphi}$. Then there exists an increasing sequence $\varphi_1 \leq \varphi_2 \leq \dots$ in Φ such that $\varphi^* = \lim_{n \rightarrow \infty} \varphi_n$ $\mathbb{P}$ -almost surely.

Definition 2.11. The random variable φ^* in Theorem 2.10 is called the *essential supremum* of Φ with respect to $\mathbb{P}$, and is denoted by

$$\text{ess sup } \Phi = \text{ess sup}_{\varphi \in \Phi} \varphi := \varphi^*.$$

The *essential infimum* of Φ with respect to $\mathbb{P}$ is defined as

$$\text{ess inf } \Phi = \text{ess inf}_{\varphi \in \Phi} \varphi := -\text{ess sup}_{\varphi \in \Phi} (-\varphi).$$

We conclude this section by recalling the notion of a generated σ -field. Let $X = (X_1, \dots, X_d) : \Omega \rightarrow \mathbb{R}^d$ be a random vector, and denote by $\mathcal{B}(\mathbb{R})$ the Borel σ -algebra. The σ -field generated by X is given by

$$\sigma(X) := \left\{ \bigcap_{i=1}^d X_i^{-1}(A) : A \in \mathcal{B}(\mathbb{R}) \right\}.$$

The following result is a version of the Doob-Dynkin lemma. It is of particular importance for the theory developed in Chapter 5. We refer to Kallenberg [13, Lemma 1.14] for its proof.

Lemma 2.12. *Let $Y : \Omega \rightarrow \mathbb{R}^n$, $n \in \mathbb{N}$, be measurable with respect to $\sigma(X)$. Then there exists some measurable function $f : \mathbb{R}^d \rightarrow \mathbb{R}^n$ such that $Y = f(X)$.*

2.2 Discrete-time financial market model

The goal of this section is to introduce the finite-discrete-time financial market model on a general probability space $(\Omega, \mathcal{F}, \mathbb{P})$. We define asset prices, trading strategies, and the concept of arbitrage. For the results presented in the following, we mainly rely on Chapters 5.1 and 5.2 of Föllmer and Schied [9]. The financial market model below is used throughout the rest of our work.

Consider a finite time horizon $T > 0$ and trading dates $t = 0, 1, \dots, T$. We work on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ endowed with a filtration $\mathbb{F} = (\mathcal{F}_t)_{t=0, \dots, T}$, representing the evolution of the available information in time. Thereby, we assume that $\mathcal{F}_T = \mathcal{F}$ and that $\mathcal{F}_0$ is trivial. We consider N assets whose price fluctuation is modeled by a nonnegative adapted $\mathbb{R}^N$ -valued stochastic process

$$\bar{S} = (S^0, S) = (S_t^0, S_t^1, \dots, S_t^d)_{t=0, \dots, T},$$

where $d = N - 1 \geq 1$ and $N, d \in \mathbb{N}$.

In our model, we assume that S^0 is a *numéraire (asset)*, meaning that $S_t^0 > 0$ $\mathbb{P}$ -almost surely for all $t = 0, \dots, T$. The numéraire acts as a unit of account, in the sense that it allows us to define the (S^0) -discounted price of the i -th asset at time t by

$$X_t^i := \frac{S_t^i}{S_t^0}, \quad \text{for } t = 0, \dots, T, i = 0, \dots, d.$$

Thus, the *discounted price process* is given by

$$\bar{X} = (X^0, X) = (X_t^0, X_t^1, \dots, X_t^d)_{t=0, 1, \dots, T},$$

where $X^0 \equiv 1$ by definition and the time- t price of the remaining d assets $X_t = (X_t^1, \dots, X_t^d)$ is expressed in units of the numéraire. We note that the process $\bar{X}$ is also nonnegative and adapted by definition. The assumption of having a numéraire asset is nontrivial, as it allows the transfer of wealth over time. Indeed, since $S_t^0 > 0$ $\mathbb{P}$ -almost surely for all t , the wealth invested in the numéraire never vanishes.

Definition 2.13. A *trading strategy* is a predictable $\mathbb{R}^N$ -valued process

$$\bar{\xi} = (\xi^0, \xi) = (\xi_t^0, \xi_t^1, \dots, \xi_t^d)_{t=1, \dots, T}.$$

We note that any trading strategy $\bar{\xi}$ starts at $t = 1$. Indeed, ξ_t^i corresponds to the number of units of the i -th asset held during the t -th trading period, that is, chosen at trading date $t - 1$ and held up to t . It is therefore natural that any trading strategy should be predictable, since we think of $\mathcal{F}_{t-1}$ as the information available at time $t - 1$, for $t = 1, \dots, T$.

Definition 2.14. Let $\bar{\xi} = (\xi^0, \xi)$ be a trading strategy.

(a) The *(discounted) value process* $V = (V_t)_{t=0, \dots, T}$ of $\bar{\xi}$ is defined as

$$V_0 := \bar{\xi}_1 \cdot \bar{X}_0 \quad \text{and} \quad V_t := \bar{\xi}_t \cdot \bar{X}_t, \quad t = 1, \dots, T.$$

(b) The *(discounted) gains process* $G = (G_t)_{t=0, \dots, T}$ of $\bar{\xi}$ is defined as

$$G_0 := 0 \quad \text{and} \quad G_t := \sum_{k=1}^t \xi_k \cdot (X_k - X_{k-1}), \quad t = 1, \dots, T.$$

Here, V_t represents the value of the portfolio $\bar{\xi}_t$ at the end of the t -th trading period in units of the numéraire, and G_t reflects the cumulative net gains of the trading strategy $\bar{\xi}$ up to time t in units of the numéraire.

Remark 2.15. We sometimes use the notation

$$\Delta X_k := (X_k - X_{k-1}), \quad k = 1, \dots, T.$$

Moreover, the gains process G of a trading strategy $\bar{\xi} = (\xi^0, \xi)$ coincides with the (discrete) *stochastic integral* of ξ with respect to X , given by

$$\left(\sum_{k=1}^t \xi_k \cdot \Delta X_k \right)_{t=0, \dots, T}$$

(with the convention that the empty sum equals 0).

Definition 2.16. A trading strategy $\bar{\xi}$ is called *self-financing* if

$$(\bar{\xi}_{t+1} - \bar{\xi}_t) \cdot \bar{S}_t = 0, \quad \text{for } t = 1, \dots, T - 1. \quad (2.3)$$

We note that $\bar{\xi}_t \cdot \bar{S}_t$ corresponds to the total value of the portfolio $\bar{\xi}_t$ at time t , whereas $\bar{\xi}_{t+1} \cdot \bar{S}_t$ is the value invested in the portfolio $\bar{\xi}_{t+1}$ at time t . In other words, (2.3) implies that the net cost or revenue of adjusting the portfolio should always be zero. The following result provides a useful characterization of a self-financing trading strategy via its value process.

Proposition 2.17. *A trading strategy $\bar{\xi}$ is self-financing if and only if its value process V satisfies*

$$V_t = V_0 + G_t = \bar{\xi}_1 \cdot \bar{X}_0 + \sum_{k=1}^t \xi_k \cdot (X_k - X_{k-1}) \quad \mathbb{P}\text{-a.s.}, \text{ for } t = 0, \dots, T.$$

Proof. Recall first that $\bar{X}_t = \frac{\bar{S}_t}{S_t^0}$ by definition. Since $S_t^0 > 0$ $\mathbb{P}$ -a.s. for all t , dividing both sides of (2.3) by S_t^0 implies that $\bar{\xi}$ is self-financing if and only if

$$(\bar{\xi}_{t+1} - \bar{\xi}_t) \cdot \bar{X}_t = 0 \quad \mathbb{P}\text{-a.s.}, \text{ for } t = 1, \dots, T-1. \quad (2.4)$$

By adding and subtracting $\bar{\xi}_{t+1} \cdot \bar{X}_{t+1}$, we see that (2.4) can be rewritten as

$$\begin{aligned} 0 &= (\bar{\xi}_{t+1} - \bar{\xi}_t) \cdot \bar{X}_t \\ &= \bar{\xi}_{t+1} \cdot \bar{X}_{t+1} - \bar{\xi}_t \cdot \bar{X}_t - \bar{\xi}_{t+1} \cdot \bar{X}_{t+1} + \bar{\xi}_{t+1} \cdot \bar{X}_t \\ &= V_{t+1} - V_t - \xi_{t+1}^0 X_{t+1}^0 + \xi_{t+1}^0 X_t^0 - \xi_{t+1} \cdot (X_{t+1} - X_t) \\ &= V_{t+1} - V_t - \xi_{t+1} \cdot (X_{t+1} - X_t) \end{aligned} \quad (2.5)$$

$\mathbb{P}$ -almost surely for $t = 1, \dots, T-1$, where in the last equation we used that $X^0 \equiv 1$. Moreover, (2.5) also holds for $t = 0$ since by definition,

$$V_1 - V_0 = \bar{\xi}_1 \cdot \bar{X}_1 - \bar{\xi}_1 \cdot \bar{X}_0 = \xi_1 \cdot (X_1 - X_0).$$

Therefore, for any $t = 1, \dots, T$, summation of (2.5) yields

$$0 = \sum_{k=1}^t (V_k - V_{k-1} - \xi_k \cdot (X_k - X_{k-1})) = V_t - V_0 - G_t \quad \mathbb{P}\text{-a.s.}$$

Since $G_0 = 0$, this proves the claim for all $t = 0, \dots, T$. $\square$

We note that since $V_0 = \bar{\xi}_1 \cdot \bar{X}_0$ is $\mathcal{F}_0$ -measurable, it is a constant, which can be interpreted as the initial investment for the purchase of the portfolio $\bar{\xi}_1$ in units of the numéraire. Proposition 2.17 then asserts that up to this constant, all changes of the value process V are caused by fluctuations of the asset prices, i.e., changes of the gains process G .

Remark 2.18. Expanding (2.4) above and recalling that $X^0 \equiv 1$, we see that the numéraire component ξ^0 of any self-financing strategy $\bar{\xi} = (\xi^0, \xi)$ with value process V satisfies

$$0 = (\xi_{t+1}^0 - \xi_t^0) X_t^0 + (\xi_{t+1} - \xi_t) \cdot X_t = (\xi_{t+1}^0 - \xi_t^0) + (\xi_{t+1} - \xi_t) \cdot X_t,$$

or equivalently

$$\xi_{t+1}^0 = \xi_t^0 - (\xi_{t+1} - \xi_t) \cdot X_t, \quad (2.6)$$

$\mathbb{P}$ -almost surely, for $t = 1, \dots, T-1$. Moreover, by definition we have

$$V_0 = \bar{\xi}_1 \cdot \bar{X}_0 = \xi_1^0 X_0^0 + \xi_1 \cdot X_0 = \xi_1^0 + \xi_1 \cdot X_0.$$

Therefore, given any constant $c \in \mathbb{R}$, representing the initial investment, and any d -dimensional predictable process ξ , by letting

$$\xi_1^0 := c - \xi_1 \cdot X_0, \quad (2.7)$$

and recursively defining ξ_t^0 through (2.6) for $t = 2, \dots, T$, (ξ^0, ξ) yields a self-financing trading strategy. Thus, it suffices to know the initial investment c and the d -dimensional predictable process ξ to describe a self-financing trading strategy.

Denote the set of d -dimensional predictable processes by

$$\Xi = \{ \xi = (\xi_t)_{t=1, \dots, T} : \xi_t : \Omega \rightarrow \mathbb{R}^d \text{ is } \mathcal{F}_{t-1}\text{-measurable for } t = 1, \dots, T \}.$$

Motivated by Remark 2.18, throughout the rest of our work, for $c \in \mathbb{R}$ and $\xi \in \Xi$, we identify the pair (c, ξ) with a self-financing trading strategy $\bar{\xi} = (\xi^0, \xi)$, with ξ^0 given as in equations (2.6) and (2.7).

For the final part of this section, we introduce the reader to the basics of arbitrage theory. An arbitrage opportunity can be interpreted as an investment strategy that allows for profit without any downside risk. As such, it should be avoided in any reasonable market model.

Definition 2.19. An *arbitrage opportunity* is a self-financing trading strategy, whose value process V satisfies

- (a) $V_0 \leq 0$
- (b) $V_T \geq 0$ $\mathbb{P}$ -a.s.
- (c) $\mathbb{P}[V_T > 0] > 0$.

A market in which there exist no arbitrage opportunities is called *arbitrage-free*.

Definition 2.20. A probability measure Q on $\mathcal{F} = \mathcal{F}_T$ is called a *martingale measure* if the discounted price process X is a (d -dimensional) Q -martingale, i.e., for each $t = 1, \dots, T, i = 1, \dots, d$,

$$\mathbb{E}^Q[X_t^i] < \infty \quad \text{and} \quad \mathbb{E}^Q[X_t^i | \mathcal{F}_{t-1}] = X_{t-1}^i \quad Q\text{-a.s.}$$

A martingale measure Q is called an *equivalent martingale measure* if it is equivalent on $\mathcal{F}$ to the probability measure $\mathbb{P}$, i.e., for any set $A \in \mathcal{F}$, $\mathbb{P}[A] = 0$ if and only if $Q[A] = 0$. In this case, we also write $Q \approx \mathbb{P}$. The set of all equivalent martingale measures is denoted by $\mathcal{P} = \mathcal{P}(X, \mathbb{P})$.

Next, we present a characterization of equivalent martingale measures as in Föllmer and Schied [9, Theorem 5.14]. The absolute value of the negative part of a random variable Y is denoted by $Y^- := -\min(0, Y)$. The proof of Theorem 2.21 is given in the Appendix.

Theorem 2.21. *Let Q be a probability measure on $\mathcal{F}$ such that $Q \approx \mathbb{P}$. Then the following are equivalent.*

- (i) *Q is an equivalent martingale measure, i.e., $Q \in \mathcal{P}$.*
- (ii) *If $\bar{\xi} = (\xi^0, \xi)$ is a self-financing strategy and ξ is bounded, then the value process V of $\bar{\xi}$ is a Q -martingale.*
- (iii) *If $\bar{\xi} = (\xi^0, \xi)$ is a self-financing strategy and its value process V satisfies $\mathbb{E}^Q[V_T^-] < \infty$, then V is a Q -martingale.*
- (iv) *If $\bar{\xi} = (\xi^0, \xi)$ is a self-financing strategy and its value process V satisfies $V_T \geq 0$ Q -a.s., then $\mathbb{E}^Q[V_T] = V_0$.*

Remark 2.22. In particular, the theorem above shows that the strong assumption $\mathbb{P} \in \mathcal{P}$ is a sufficient condition for absence of arbitrage in the market. Indeed, if $\bar{\xi}$ is a self-financing trading strategy with $V_0 \leq 0$ and $V_T \geq 0$ $\mathbb{P}$ -a.s. (see Definition 2.19), then by part (iv) of Theorem 2.21 we have

$$0 \leq \mathbb{E}[V_T] = V_0 \leq 0,$$

and hence $V_T = 0$ $\mathbb{P}$ -almost surely. However, the next result shows that the market being arbitrage-free is a significantly weaker condition than $\mathbb{P} \in \mathcal{P}$.

The following theorem is one of the most important results of mathematical finance, called *(first) fundamental theorem of asset pricing*. It provides a characterization of an arbitrage-free market via the set of equivalent martingale measures $\mathcal{P}$. For a finite-discrete-time setting the theorem was first proved by Dalang *et al.* [3]. Throughout the remaining part of this thesis, we develop our theory for an arbitrage-free market model. Assumption 2.24 below introduces this standing assumption.

Theorem 2.23 (First fundamental theorem of asset pricing). *The market model is arbitrage-free if and only if the set $\mathcal{P}$ of all equivalent martingale measures is nonempty.*

Assumption 2.24. The set of all equivalent martingale measures $\mathcal{P}(X, \mathbb{P})$ is nonempty.

Definition 2.25. Let $a \geq 0$ and let $\bar{\xi}$ be a self-financing trading strategy. We call $\bar{\xi}$ *a-admissible* if its value process V satisfies

$$V_t \geq -a, \quad \mathbb{P}\text{-a.s., for } t = 0, \dots, T.$$

We denote the set of all a -admissible strategies by $\mathcal{A}^a$. The strategy $\bar{\xi}$ is called *admissible*, if there exists $a \geq 0$ such that $\bar{\xi}$ is a -admissible.

Admissibility of a trading strategy ensures that the value of the portfolio does not go below a certain lower bound. This represents a reasonable requirement, as normally there is a limit to the amount of money participants of the market have at their disposal. In particular, 0-admissibility can be interpreted as a guarantee that the buyer of the strategy will be in debt at no time t .

Remark 2.26. In an arbitrage-free market, to check admissibility of a strategy it suffices to look at the final value of the portfolio. Indeed, let $\bar{\xi} = (\xi^0, \xi)$ be a self-financing trading strategy with value process V , and suppose that

$$V_T \geq -a \quad \mathbb{P}\text{-a.s.}, \text{ for some } a \geq 0. \quad (2.8)$$

We show that $\bar{\xi}$ is a -admissible. Suppose for contradiction that

$$\mathbb{P}[V_{t^*} < -a] > 0, \quad \text{for some } t^* \in \{0, \dots, T-1\}.$$

Let $A = \{\omega \in \Omega : V_{t^*}(\omega) < -a\}$. By (2.8) and Proposition 2.17, we have

$$-a \leq V_T = V_0 + \sum_{k=1}^T \xi_k \cdot \Delta X_k = V_{t^*} + \sum_{k=t^*+1}^T \xi_k \cdot \Delta X_k \quad \mathbb{P}\text{-a.s.}$$

Equivalently, by definition of A we get

$$\mathbb{1}_A \sum_{k=t^*+1}^T \xi_k \cdot \Delta X_k \geq -a\mathbb{1}_A - V_{t^*}\mathbb{1}_A \geq 0 \quad \mathbb{P}\text{-a.s.}, \quad (2.9)$$

with the last inequality being strict on A . For $s = 0, \dots, T$, define the d -dimensional process

$$\tilde{\xi}_s := \begin{cases} 0 & \text{for } s \leq t^* \\ \mathbb{1}_A \xi_s & \text{for } s > t^*. \end{cases}$$

We note that $\tilde{\xi}$ is predictable, since $\bar{\xi}$ is predictable and A is $\mathcal{F}_{t^*}$ -measurable. Thus we can construct a self-financing trading strategy $(\tilde{V}_0, \tilde{\xi})$ with initial investment $\tilde{V}_0 = 0$, as in Remark 2.18. Let $\tilde{V}$ denote the value process of $\tilde{\xi}$. Then, by (2.9) we get

$$\tilde{V}_T = 0 + \sum_{k=1}^T \tilde{\xi}_k \cdot \Delta X_k = \sum_{k=t^*+1}^T \mathbb{1}_A (\xi_k \cdot \Delta X_k) \geq 0 \quad \mathbb{P}\text{-a.s.},$$

with strict inequality on A . Hence we conclude

$$0 < \mathbb{P}[V_{t^*} < -a] = \mathbb{P}[A] \leq \mathbb{P}[\tilde{V}_T > 0],$$

so $(\tilde{V}_0, \tilde{\xi})$ is an arbitrage opportunity, yielding a contradiction.

Chapter 3

Superhedging

In this chapter, we introduce the concept of superhedging. In an incomplete market, for the seller of a discounted European contingent claim it may be impossible to follow a self-financing trading strategy whose final value coincides with the payoff of the claim. Superhedging strategies provide an alternative, ensuring the seller stays on the safe side. Section 3.1 is devoted to the pricing of a European contingent claim H , with an emphasis on the difference between complete and incomplete markets. In Section 3.2 we derive the superhedging price of H at time $t = 0$ using the optional decomposition theorem. We then extend this to a comparable result in Section 3.3, characterizing the superhedging price process for times $t > 0$. The content of this section mainly relies on Chapters 7.1–7.3 of Föllmer and Schied [9], with the difference that we focus on European contingent claims only.

3.1 European contingent claims

Contingent claims are derivative instruments whose payoff depends on the value of an underlying asset. European contingent claims are characterized by the fact that they can only be redeemed at the maturity date $T > 0$ of the contract, in contrast to American contingent claims, whose buyer can exercise the claim at any time $t = 0, \dots, T$. We recall that by Assumption 2.24,

$$\mathcal{P}(X, \mathbb{P}) \neq \emptyset,$$

or in other words, we consider an arbitrage-free market model. This section introduces the pricing of a European contingent claim H . In particular, we define the notion of arbitrage-free prices and describe the set of these prices for H , both in complete and incomplete market models.

In the case of our financial market model from Section 2.2, the underlying assets are the primary assets $S^0, \dots, S^d$. A European contingent claim is modeled by a nonnegative random variable C on $(\Omega, \mathcal{F}_T, \mathbb{P})$, where we recall that $\mathcal{F} = \mathcal{F}_T$ by assumption. In other words, C represents an asset that has payoff $C(\omega)$ at time T , for some market state $\omega \in \Omega$. Given the assumption of a numéraire asset S^0 in our market model, for the purpose of this work we

only focus on discounted European contingent claims given by

$$H := \frac{C}{S_T^0},$$

for some European contingent claim C . Moreover, we assume that any discounted European contingent claim H satisfies the following condition.

Definition 3.1. A (discounted) *European contingent claim* is a nonnegative random variable H on $(\Omega, \mathcal{F}, \mathbb{P})$ such that

$$\sup_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[H] < \infty. \quad (3.1)$$

Throughout this work, we only consider European contingent claims as in the definition above, i.e., satisfying (3.1). We sometimes just say *contingent claim* or *claim*. Now we turn to the pricing of a European contingent claim H . Hereby, one wants to make sure that by adding H as an asset into the market model $(\Omega, \mathcal{F}, \mathbb{F}, \mathbb{P}, X)$, no new arbitrage opportunities arise. The next definition formalizes this idea.

Definition 3.2. Let H be a discounted European contingent claim. An *arbitrage-free price* of H is a real number $\pi^H \geq 0$, such that there exists a nonnegative real-valued adapted process $X^{d+1} = (X_t^{d+1})_{t=0, \dots, T}$ with

$$X_0^{d+1} = \pi^H \quad \text{and} \quad X_T^{d+1} = H \quad \mathbb{P}\text{-a.s.},$$

and such that the enlarged market model with price process $(\bar{X}, X^{d+1})$ is arbitrage-free. We denote by $\Pi(H)$ the set of all arbitrage-free prices of H , and by

$$\pi_{\inf}(H) := \inf \Pi(H), \quad \pi_{\sup}(H) := \sup \Pi(H)$$

the lower and the upper bounds of $\Pi(H)$, respectively.

Remark 3.3. We note that any arbitrage-free price π^H is given in units of the numéraire S^0 .

Theorem 3.4. Let H be a discounted European contingent claim. Then the set of arbitrage-free prices of H is nonempty and given by

$$\Pi(H) = \{\mathbb{E}^*[H] : \mathbb{P}^* \in \mathcal{P}\}. \quad (3.2)$$

Remark 3.5. We recall that since $\sup_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[H] < \infty$ by assumption, we have $\mathbb{E}^*[H] < \infty$ for all equivalent martingale measures $\mathbb{P}^* \in \mathcal{P}$. In particular, by the theorem above condition (3.1) on the contingent claim H implies that

$$\pi_{\sup}(H) < \infty.$$

Proof of Theorem 3.4. To show (3.2), suppose first that π^H is an arbitrage-free price of H . Let X^{d+1} be a nonnegative adapted process as in Definition 3.2. Then, by Theorem 2.23 there exists $\bar{\mathbb{P}} \in \mathcal{P}$ for the market model extended by X^{d+1} . In particular, (X, X^{d+1}) is a $\bar{\mathbb{P}}$ -martingale and therefore

$$\pi^H = X_0^{d+1} = \bar{\mathbb{E}}[X_T^{d+1} | \mathcal{F}_0] = \bar{\mathbb{E}}[H].$$

This shows inclusion “ $\subseteq$ ” in (3.2).

For the other inclusion, let $\pi^H := \mathbb{E}^*[H]$ for some $\mathbb{P}^* \in \mathcal{P}$. We define the process

$$X_t^{d+1} := \mathbb{E}^*[H | \mathcal{F}_t], \quad t = 0, \dots, T.$$

Then X^{d+1} is clearly nonnegative and adapted, and we have

$$X_0^{d+1} = \mathbb{E}^*[H | \mathcal{F}_0] = \mathbb{E}^*[H] = \pi^H, \quad X_T^{d+1} = \mathbb{E}^*[H | \mathcal{F}_T] = H \quad \mathbb{P}^*\text{-a.s.} \quad (3.3)$$

Moreover, by the tower property

$$\mathbb{E}^*[X_t^{d+1} | \mathcal{F}_{t-1}] = \mathbb{E}^*[\mathbb{E}^*[H | \mathcal{F}_t] | \mathcal{F}_{t-1}] = \mathbb{E}^*[H | \mathcal{F}_{t-1}] = X_{t-1}^{d+1}$$

for $t = 1, \dots, T$, so (X, X^{d+1}) is a $\mathbb{P}^*$ -martingale. Thus, the same probability measure $\mathbb{P}^*$ is also an equivalent martingale measure for the market model enlarged by X^{d+1} , which by Theorem 2.23 implies that the enlarged market is arbitrage-free. Combined with (3.3), this implies $\pi^H \in \Pi(H)$, and since $\mathbb{P}^* \in \mathcal{P}$ was arbitrary, (3.2) follows. Then recalling that $\mathcal{P}(X, \mathbb{P}) \neq \emptyset$ by Assumption 2.24, it follows that the set $\Pi(H)$ is nonempty, which concludes the proof. $\square$

Definition 3.6. A European contingent claim H is called *attainable* if there exists a self-financing trading strategy $\bar{\xi}$ with value process V such that

$$H = \bar{\xi}_T \cdot \bar{X}_T = V_T \quad \mathbb{P}\text{-a.s.}$$

In this case, $\bar{\xi}$ is called a *replicating strategy* for H .

Definition 3.7. An arbitrage-free market in which every contingent claim H is attainable is called *complete*.

At the expiration of the contract at time T , the seller of a European contingent claim H must pay the amount $H(\omega)$, depending on the market state $\omega \in \Omega$. If H is attainable, by purchasing a replicating strategy $\bar{\xi}$ at time 0, the seller is able to guarantee the payoff of the claim H , thus eliminating any risk. We emphasize that our work is in a general incomplete market setting. However, for the sake of a greater understanding, we now state two important results for the characterization of complete markets. For the proofs of the theorems below, we refer the reader to Föllmer and Schied [9, Theorems 5.32 and 5.37].

Theorem 3.8. *Let H be a European contingent claim.*

(i) *If H is attainable, then*

$$\Pi(H) = \{V_0\}, \quad (3.4)$$

where V is the value process of any replicating strategy for H .

(ii) *If H is not attainable, then $\pi_{\inf}(H) < \pi_{\sup}(H)$ and*

$$\Pi(H) = (\pi_{\inf}(H), \pi_{\sup}(H)).$$

Remark 3.9. For an attainable contingent claim H as in (i) above, all replicating strategies have the same value process (see [9, Theorem 5.25]). Thus, (3.4) is well-defined.

Theorem 3.10 (Second fundamental theorem of asset pricing). *An arbitrage-free market model is complete if and only if there exists a unique equivalent martingale measure $\mathbb{P}^* \in \mathcal{P}$.*

Completeness of the market therefore simplifies the pricing of a contingent claim H significantly, as H admits a unique arbitrage-free price. Part (i) of Theorem 3.8 shows that while following a replication strategy eliminates any risk, it also stops the seller from making a profit. Indeed, in this case, the unique arbitrage-free price of H coincides with the initial cost of any replicating strategy. In an incomplete market model, where not all claims are attainable, superhedging provides an alternative approach.

3.2 Superhedging price at time $t = 0$

We previously saw that in an incomplete market model not every contingent claim can be perfectly replicated. The idea of superhedging is to avoid any risk for the seller, by following a self-financing strategy whose final value covers the payoff of the claim. In this section, we show that for any European contingent claim H such a strategy exists. Moreover, we determine the minimal initial investment needed for its purchase, that is, the superhedging price of H at time 0.

Definition 3.11. Let H be a European contingent claim. A *superhedging strategy* for H is a self-financing trading strategy $\bar{\xi}$ whose value process V satisfies

$$V_T \geq H \quad \mathbb{P}\text{-a.s.}$$

Remark 3.12. Note that since our market model is arbitrage-free, any superhedging strategy $\bar{\xi}$ for a contingent claim H is automatically 0-admissible by the nonnegativity of H and by Remark 2.26.

In other words, a superhedging strategy allows the seller of the claim to avoid any risk, by keeping the final value of the portfolio above the strike price. For a European contingent claim H , we define the set

$$\begin{aligned}\mathcal{U}_0(H) &:= \left\{ c \in \mathbb{R} : \exists \xi \in \Xi \text{ s.t. } c + \sum_{k=1}^T \xi_k \cdot (X_k - X_{k-1}) \geq H \text{ } \mathbb{P}\text{-a.s.} \right\} \\ &= \left\{ c \in \mathbb{R} : \exists \xi \in \Xi \text{ s.t. } (c, \xi) \text{ is a superhedging strategy for } H \right\},\end{aligned}$$

where we recall that the second equality follows from Remark 2.18. In the following, our goal is to find the minimal initial investment needed to purchase a superhedging strategy for H , that is, $\inf \mathcal{U}_0(H)$.

Remark 3.13. We observe that the set $\mathcal{U}_0$ is either empty, or has the form of an interval

$$(\inf \mathcal{U}_0, +\infty) \quad \text{or} \quad [\inf \mathcal{U}_0, +\infty),$$

depending on whether the infimum is attained in $\mathcal{U}_0$.

Definition 3.14. An adapted process $(M_t)_{t=0,\dots,T}$ is called a $\mathcal{P}$ -martingale if it is a $\mathbb{P}^*$ -martingale for each $\mathbb{P}^* \in \mathcal{P}$. We define a $\mathcal{P}$ -supermartingale and a $\mathcal{P}$ -submartingale analogously.

Definition 3.15. Let H be a European contingent claim. The *upper Snell envelope* of H is defined as

$$U_t^\uparrow(H) = U_t^\uparrow := \operatorname{ess\,sup}_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[H|\mathcal{F}_t], \quad \text{for } t = 0, \dots, T.$$

Remark 3.16. We note that since $\mathcal{F}_0$ is trivial and $\mathcal{F}_T = \mathcal{F}$, we have

$$U_0^\uparrow = \sup_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[H] < \infty \quad \text{and} \quad U_T^\uparrow = H \text{ } \mathbb{P}\text{-a.s.}$$

Lemma 3.17. The upper Snell envelope $U^\uparrow(H)$ of a European contingent claim H satisfies the recursive scheme

$$U_t^\uparrow = \operatorname{ess\,sup}_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[U_{t+1}^\uparrow|\mathcal{F}_t] \quad \mathbb{P}\text{-a.s., for } t = T-1, \dots, 0. \quad (3.5)$$

Proof. Fix $t \in \{0, \dots, T-1\}$. By definition of $U_t^\uparrow$ and using the tower property, we have

$$\begin{aligned}U_t^\uparrow &= \operatorname{ess\,sup}_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[H|\mathcal{F}_t] = \operatorname{ess\,sup}_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[\mathbb{E}^*[H|\mathcal{F}_{t+1}]|\mathcal{F}_t] \\ &\leq \operatorname{ess\,sup}_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[\operatorname{ess\,sup}_{\mathbb{P}^{**} \in \mathcal{P}} \mathbb{E}^{**}[H|\mathcal{F}_{t+1}]|\mathcal{F}_t] = \operatorname{ess\,sup}_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[U_{t+1}^\uparrow|\mathcal{F}_t] \quad \mathbb{P}\text{-a.s.},\end{aligned}$$

which proves inequality “ $\leq$ ” of (3.5). For the converse inequality, fix $\tilde{\mathbb{P}} \in \mathcal{P}$. We define

$$\mathcal{P}_{t+1}(\tilde{\mathbb{P}}) := \{\mathbb{P}^* \in \mathcal{P} : \mathbb{P}^*[A] = \tilde{\mathbb{P}}[A] \text{ for all } A \in \mathcal{F}_{t+1}\},$$

denoting the set of equivalent martingale measures which coincide with $\tilde{\mathbb{P}}$ on $\mathcal{F}_{t+1}$. Consider the set

$$\Phi := \{\mathbb{E}^*[H|\mathcal{F}_{t+1}] : \mathbb{P}^* \in \mathcal{P}_{t+1}(\tilde{\mathbb{P}})\}.$$

Then, by definition of the conditional expectation we have

$$\text{ess sup}_{\mathbb{P}^* \in \mathcal{P}} \Phi = \text{ess sup}_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[H|\mathcal{F}_{t+1}] = U_{t+1}^\uparrow \quad \mathbb{P}\text{-a.s.}$$

Moreover, Φ is directed upwards, so by part (a) of Theorem 2.10 there exists an increasing sequence $(\mathbb{P}^{(k)})_{k \in \mathbb{N}} \subset \mathcal{P}_{t+1}(\tilde{\mathbb{P}})$ such that

$$\lim_{k \rightarrow \infty} \mathbb{E}^{(k)}[H|\mathcal{F}_{t+1}] = \text{ess sup}_{\mathbb{P}^* \in \mathcal{P}} \Phi = U_{t+1}^\uparrow \quad \mathbb{P}\text{-a.s.}$$

Using Fatou's lemma for conditional expectations, we obtain

$$\begin{aligned} \tilde{\mathbb{E}}[U_{t+1}^\uparrow | \mathcal{F}_t] &= \tilde{\mathbb{E}}\left[\lim_{k \rightarrow \infty} \mathbb{E}^{(k)}[H|\mathcal{F}_{t+1}] | \mathcal{F}_t\right] \leq \liminf_{k \rightarrow \infty} \tilde{\mathbb{E}}[\mathbb{E}^{(k)}[H|\mathcal{F}_{t+1}] | \mathcal{F}_t] \\ &\leq \text{ess sup}_{\tilde{\mathbb{P}} \in \mathcal{P}_{t+1}(\tilde{\mathbb{P}})} \tilde{\mathbb{E}}[\hat{\mathbb{E}}[H|\mathcal{F}_{t+1}] | \mathcal{F}_t] = \tilde{\mathbb{E}}[\hat{\mathbb{E}}[H|\mathcal{F}_{t+1}] | \mathcal{F}_t] \\ &\leq \text{ess sup}_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[H|\mathcal{F}_t] = U_t^\uparrow \quad \mathbb{P}\text{-a.s.}, \end{aligned}$$

where the fourth step follows from the fact that $\hat{\mathbb{E}}[H|\mathcal{F}_{t+1}] = \tilde{\mathbb{E}}[H|\mathcal{F}_{t+1}]$ $\mathbb{P}$ -a.s. for any $\tilde{\mathbb{P}} \in \mathcal{P}_{t+1}(\tilde{\mathbb{P}})$. Since $\tilde{\mathbb{P}} \in \mathcal{P}$ was arbitrary, we conclude

$$\text{ess sup}_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[U_{t+1}^\uparrow | \mathcal{F}_t] \leq U_t^\uparrow,$$

which proves (3.5). Since $t \in \{0, \dots, T-1\}$ was arbitrary, this shows the lemma. $\square$

Theorem 3.18. *Let H be a European contingent claim. Then the upper Snell envelope $U^\uparrow(H)$ is the smallest $\mathcal{P}$ -supermartingale whose terminal value dominates H .*

Proof. Let $\tilde{\mathbb{P}} \in \mathcal{P}$. Then, by Lemma 3.17 we have

$$U_t^\uparrow = \text{ess sup}_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[U_{t+1}^\uparrow | \mathcal{F}_t] \geq \tilde{\mathbb{E}}[U_{t+1}^\uparrow | \mathcal{F}_t] \quad \tilde{\mathbb{P}}\text{-a.s. for } t = 0, \dots, T-1,$$

which in turn implies

$$\tilde{\mathbb{E}}[U_t^\uparrow] = \tilde{\mathbb{E}}[\tilde{\mathbb{E}}[U_t^\uparrow | \mathcal{F}_{t-1}]] \leq \tilde{\mathbb{E}}[U_{t-1}^\uparrow] \leq \dots \leq U_0^\uparrow = \sup_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[H] < \infty,$$

for $t = 1, \dots, T$. Hence we conclude that $U^\uparrow(H)$ is a $\tilde{\mathbb{P}}$ -supermartingale. Since $\tilde{\mathbb{P}} \in \mathcal{P}$ was arbitrary and $U_T^\uparrow = H$ $\mathbb{P}$ -almost surely, it follows that $U^\uparrow$ is a

$\mathcal{P}$ -supermartingale whose terminal value dominates H . Finally, suppose $\tilde{U}$ is another $\mathcal{P}$ -supermartingale such that

$$\tilde{U}_T \geq H = U_T^\uparrow \quad \mathbb{P}\text{-a.s.} \quad (3.6)$$

We show by backward induction that $U_t^\uparrow \leq \tilde{U}_t$ $\mathbb{P}$ -a.s. for $t = 0, \dots, T$. Suppose $\tilde{U}_{t+1} \geq U_{t+1}^\uparrow$ $\mathbb{P}$ -a.s. for some $t \in \{T-1, \dots, 0\}$. Then, by the supermartingale property of $\tilde{U}$ and the induction hypothesis, for any $\mathbb{P}^* \in \mathcal{P}$ we have

$$\tilde{U}_t \geq \mathbb{E}^*[\tilde{U}_{t+1} \mid \mathcal{F}_t] \geq \mathbb{E}^*[U_{t+1}^\uparrow \mid \mathcal{F}_t] \quad \mathbb{P}\text{-a.s.}$$

Therefore, by the definition of the essential supremum and Lemma 3.17, we obtain

$$\tilde{U}_t \geq \operatorname{ess\,sup}_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[U_{t+1}^\uparrow \mid \mathcal{F}_t] = U_t^\uparrow \quad \mathbb{P}\text{-a.s.},$$

which shows the induction step, and together with (3.6) concludes the proof. $\square$

The following theorem provides a characterization of nonnegative $\mathcal{P}$ -supermartingales. It was first proved by El Karoui and Quenez [5, Theorem 2.1.2] in continuous time, where the price process X is the solution of a stochastic differential equation. The result was later extended by Kramkov [14, Theorem 2.1] for a locally bounded price process, and finally to the general setting by Föllmer and Kabanov [6, Theorem 1]. For the proof of Theorem 3.19 in our finite-discrete-time setting, we also refer to Föllmer and Schied [9, Theorem 7.5].

Theorem 3.19 (Optional decomposition theorem). *For an adapted nonnegative process $U = (U_t)_{t=0, \dots, T}$, the following two statements are equivalent.*

- (i) U is a $\mathcal{P}$ -supermartingale.
- (ii) There exist a d -dimensional predictable process $\xi = (\xi_t)_{t=1, \dots, T}$ and an adapted increasing process $B = (B_t)_{t=0, \dots, T}$ with $B_0 = 0$, such that

$$U_t = U_0 + \sum_{k=1}^t \xi_k \cdot (X_k - X_{k-1}) - B_t \quad \mathbb{P}\text{-a.s., for } t = 0, \dots, T. \quad (3.7)$$

Remark 3.20. Let U be a nonnegative $\mathcal{P}$ -supermartingale as in Theorem 3.19, and let ξ and B as in (3.7). We note that by Proposition 2.17 and Remark 2.18, the process V , given by

$$V_t := U_0 + \sum_{k=1}^t \xi_k \cdot (X_k - X_{k-1}) \quad \text{for } t = 0, \dots, T,$$

is the value process of a self-financing trading strategy defined by (U_0, ξ) . Moreover, by (3.7) we have

$$V_T = U_T + B_T \geq B_0 = 0 \quad \mathbb{P}\text{-a.s.}$$

Thus, by part (iii) of Theorem 2.21, V is a $\mathcal{P}$ -martingale. In other words, the optional decomposition theorem asserts that an adapted nonnegative process is $\mathcal{P}$ -supermartingale if and only if it can be written as the difference of a $\mathcal{P}$ -martingale and an increasing adapted process, null at 0.

Remark 3.21. Theorem 3.19 is sometimes also referred to as the *uniform Doob-Meyer decomposition* originating from the well-known Doob-Meyer decomposition, which states that any supermartingale V can be written as the difference of a local martingale and a unique predictable increasing process, null at 0. In contrast, the optional decomposition theorem provides a decomposition simultaneously for the whole class $\mathcal{P}$. However, the process B in (3.7) is not unique and only adapted, hence explaining the name *optional*.

Representation (3.7) is called the *optional decomposition* of the supermartingale U , and B is often referred to as the *process of consumption*. Recall that the set of initial investments needed to purchase a superhedging strategy for H is given by

$$\mathcal{U}_0(H) = \left\{ c \in \mathbb{R} : \exists \xi \in \Xi \text{ s.t. } c + \sum_{k=1}^T \xi_k \cdot (X_k - X_{k-1}) \geq H \text{ } \mathbb{P}\text{-a.s.} \right\}.$$

We know from Theorem 3.18 that the upper Snell envelope of a contingent claim H is a $\mathcal{P}$ -supermartingale, which is nonnegative by definition. Using the optional decomposition of $U^\uparrow(H)$, we can now prove the main result of this section.

Proposition 3.22. *Let H be a European contingent claim. Then the upper Snell envelope $U^\uparrow(H)$ satisfies*

$$U_0^\uparrow = \sup_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[H] \in \mathcal{U}_0(H) \quad \text{and} \quad \inf \mathcal{U}_0(H) = U_0^\uparrow.$$

Proof. Since $U^\uparrow(H)$ is a nonnegative $\mathcal{P}$ -supermartingale, by Theorem 3.19 there exist $\xi \in \Xi$ and an increasing adapted process $B = (B_t)_{t=0, \dots, T}$ with $B_0 = 0$, such that

$$U_t^\uparrow = U_0^\uparrow + \sum_{k=1}^t \xi_k \cdot (X_k - X_{k-1}) - B_t \quad \mathbb{P}\text{-a.s., for } t = 0, \dots, T.$$

Moreover,

$$V_t := U_0^\uparrow + \sum_{k=1}^t \xi_k \cdot (X_k - X_{k-1}), \quad t = 0, \dots, T,$$

is the value process of the self-financing trading strategy described by $(U_0^\uparrow, \xi)$, and it satisfies

$$V_T = U_T^\uparrow + B_T = H + B_T \geq H \quad \mathbb{P}\text{-a.s.}$$

Hence, $(U_0^\uparrow, \xi)$ is a superhedging strategy for H with initial investment $U_0^\uparrow$, which shows $U_0^\uparrow \in \mathcal{U}_0(H)$. Suppose now that $\tilde{V}$ is the value process of another superhedging strategy $\tilde{\xi}$. Then $\tilde{V}_T \geq H \geq 0$ $\mathbb{P}$ -a.s., and thus also $\mathbb{P}^*$ -a.s. for each $\mathbb{P}^* \in \mathcal{P}$. Therefore, by part (iv) of Theorem 2.21 we have

$$\mathbb{E}^*[H] \leq \mathbb{E}^*[\tilde{V}_T] = \tilde{V}_0$$

for each $\mathbb{P}^* \in \mathcal{P}$, which by definition of the supremum implies

$$U_0^\uparrow = \sup_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[H] \leq \tilde{V}_0.$$

Since $\tilde{V}$ was arbitrary, this shows that $U_0^\uparrow = \inf \mathcal{U}_0(H)$. $\square$

We call $U_0^\uparrow$ the (time-0) *superhedging price* of H . We recall that by Theorem 3.4,

$$\sup_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[H] = \pi_{\text{sup}}(H).$$

Thus, the minimal investment needed to implement a superhedging strategy for H coincides with the upper bound of the arbitrage-free prices of H . If the contingent claim H is not attainable, by Theorem 3.8, $\pi_{\text{sup}}(H)$ is not an arbitrage-free price for H . In particular, the seller of H cannot sell the claim for an arbitrage-free price as high as the minimal investment needed to hedge against H . This explains why the superhedging price can sometimes be expensive, hence leading the seller of H to take on a certain amount of risk. This is the content of the next chapter on quantile hedging.

3.3 Superhedging price at time $t > 0$

We showed in the previous section that $U_0^\uparrow(H)$ is the minimal amount needed to implement a superhedging strategy for a European contingent claim H at time $t = 0$. Our aim in this section is to prove a similar result for times $t > 0$. Thereby, an important difference is that we work with random variables instead of constants, as $\mathcal{F}_t$ need not be trivial for $t > 0$. In the second part of the section, we characterize the consumption process B from the optional decomposition of the upper Snell envelope $U^\uparrow(H)$.

Let $\Delta X_k := (X_k - X_{k-1})$, $k = 1, \dots, T$. For a European contingent claim H , similarly to $\mathcal{U}_0(H)$, we define the set

$$\mathcal{U}_t(H) := \left\{ U_t \in L^0(\Omega, \mathcal{F}_t, \mathbb{P}) : \exists \xi \in \Xi \text{ s.t. } U_t + \sum_{k=t+1}^T \xi_k \cdot \Delta X_k \geq H \text{ } \mathbb{P}\text{-a.s.} \right\},$$

for $t = 1, \dots, T$. We want to find the minimal investment needed to purchase a superhedging strategy at time t . In the following, we show that the upper Snell envelope $U^\uparrow(H)$ satisfies a result comparable to Proposition 3.22, with the difference of taking the essential infimum over a set of random variables instead of the infimum of an interval.

Theorem 3.23. *The upper Snell envelope $U^\uparrow(H)$ of a European contingent claim H satisfies*

$$(i) \ U_t^\uparrow \in \mathcal{U}_t(H),$$

$$(ii) \ U_t^\uparrow = \text{ess inf } \mathcal{U}_t(H),$$

for $t = 1, \dots, T$.

Proof. (i) We recall that by Theorem 3.18, the upper Snell envelope $U^\uparrow(H)$ is a $\mathcal{P}$ -supermartingale, nonnegative by definition. Hence, by Theorem 3.19 there exists an adapted increasing process B with $B_0 = 0$ and a d -dimensional predictable process $\xi \in \Xi$ such that

$$U_t^\uparrow = U_0^\uparrow + \sum_{k=1}^t \xi_k \cdot (X_k - X_{k-1}) - B_t \quad \mathbb{P}\text{-a.s., for } t = 1, \dots, T. \quad (3.8)$$

Recalling that $U_T^\uparrow = H$ $\mathbb{P}$ -a.s., (3.8) and the fact that B is increasing yield

$$\begin{aligned} U_t^\uparrow + \sum_{k=t+1}^T \xi_k \cdot (X_k - X_{k-1}) &= U_0^\uparrow + \sum_{k=1}^T \xi_k \cdot (X_k - X_{k-1}) - B_t \\ &\geq U_0^\uparrow + \sum_{k=1}^T \xi_k \cdot (X_k - X_{k-1}) - B_T \\ &= U_T^\uparrow \\ &= H \quad \mathbb{P}\text{-a.s., for } t = 1, \dots, T. \end{aligned} \quad (3.9)$$

Thus, we conclude $U_t^\uparrow \in \mathcal{U}_t(H)$ for $t = 1, \dots, T$.

(ii) Fix $t \in \{1, \dots, T\}$. From (i) and from the definition of the essential infimum it follows immediately that $U_t^\uparrow \geq \text{ess inf } \mathcal{U}_t(H)$ $\mathbb{P}$ -a.s. For the converse inequality, let $\tilde{U}_t \in \mathcal{U}_t(H)$ be arbitrary and let $\tilde{\xi} \in \Xi$ be such that

$$\tilde{U}_t + \sum_{k=t+1}^T \tilde{\xi}_k \cdot (X_k - X_{k-1}) \geq H \quad \mathbb{P}\text{-a.s.} \quad (3.10)$$

We show that $U_t^\uparrow \leq \tilde{U}_t$ $\mathbb{P}$ -a.s. or equivalently,

$$\hat{U}_t := U_t^\uparrow \wedge \tilde{U}_t = U_t^\uparrow \quad \mathbb{P}\text{-a.s.} \quad (3.11)$$

By definition, we have $\hat{U}_t \leq U_t^\uparrow$. Let $\xi \in \Xi$ be a predictable process as in (3.8) and (3.9), and define $A := \{\omega \in \Omega : U_t^\uparrow(\omega) \leq \tilde{U}_t(\omega)\}$. Then we have $\hat{U}_t = U_t^\uparrow \mathbb{1}_A + \tilde{U}_t \mathbb{1}_{A^C}$. For $s = 0, \dots, T$, define

$$\hat{\xi}_s := \begin{cases} \xi_s & \text{if } s \leq t, \\ \xi_s \mathbb{1}_A + \tilde{\xi}_s \mathbb{1}_{A^C} & \text{if } s > t. \end{cases}$$

Since $\xi, \tilde{\xi} \in \Xi$ and A is $\mathcal{F}_t$ -measurable, we have $\hat{\xi} \in \Xi$. Moreover, by (3.9) and (3.10), and by the definitions of $\hat{U}_t$ and $\hat{\xi}$, we obtain

$$\begin{aligned} H &\leq \left(U_t^\uparrow + \sum_{k=t+1}^T \xi_k \cdot \Delta X_k \right) \mathbb{1}_A + \left(\tilde{U}_t + \sum_{k=t+1}^T \tilde{\xi}_k \cdot \Delta X_k \right) \mathbb{1}_{A^c} \\ &= U_t^\uparrow \mathbb{1}_A + \tilde{U}_t \mathbb{1}_{A^c} + \sum_{k=t+1}^T (\xi_k \mathbb{1}_A + \tilde{\xi}_k \mathbb{1}_{A^c}) \cdot \Delta X_k \\ &= \hat{U}_t + \sum_{k=t+1}^T \hat{\xi}_k \cdot \Delta X_k \quad \mathbb{P}\text{-a.s.}, \end{aligned}$$

which implies $\hat{U}_t \in \mathcal{U}_t(H)$. We define the process $\hat{V}$ by

$$\hat{V}_s := U_0^\uparrow + \sum_{k=1}^s \hat{\xi}_k \cdot (X_k - X_{k-1}), \quad \text{for } s = 0, \dots, T.$$

Then $\hat{V}$ is the value process of a self-financing trading strategy defined by $(U_0^\uparrow, \hat{\xi})$. Moreover, by the definition of $\hat{\xi}$ and (3.8) we have

$$\begin{aligned} \hat{V}_T &= U_0^\uparrow + \sum_{k=1}^T \hat{\xi}_k \cdot (X_k - X_{k-1}) \\ &= U_0^\uparrow + \sum_{k=1}^t \xi_k \cdot (X_k - X_{k-1}) + \sum_{k=t+1}^T \hat{\xi}_k \cdot (X_k - X_{k-1}) \\ &\geq U_t^\uparrow + \sum_{k=t+1}^T \hat{\xi}_k \cdot (X_k - X_{k-1}) \\ &\geq \hat{U}_t + \sum_{k=t+1}^T \hat{\xi}_k \cdot (X_k - X_{k-1}) \geq H \geq 0 \quad \mathbb{P}\text{-a.s.}, \end{aligned}$$

By part (iii) of Theorem 2.21, it follows that $\hat{V}$ is a $\mathcal{P}$ -martingale. Hence, we obtain

$$\begin{aligned} U_t^\uparrow &= \text{ess sup}_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[H | \mathcal{F}_t] \\ &\leq \text{ess sup}_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[\hat{U}_t + \sum_{k=t+1}^T \hat{\xi}_k \cdot (X_k - X_{k-1}) | \mathcal{F}_t] \\ &= \text{ess sup}_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[\hat{U}_t + \hat{V}_T - \hat{V}_t | \mathcal{F}_t] \\ &= \hat{U}_t \quad \mathbb{P}\text{-a.s.}, \end{aligned}$$

which proves (3.11) and thus shows $U_t^\uparrow \leq \tilde{U}_t$ $\mathbb{P}$ -a.s. Since $\tilde{U}_t \in \mathcal{U}_t$ was arbitrary, by definition of the essential infimum we conclude

$$U_t^\uparrow \leq \text{ess inf } \mathcal{U}_t(H) \quad \mathbb{P}\text{-a.s.}$$

As $t \in \{1, \dots, T\}$ was arbitrary, this shows (ii) and hence concludes the proof. $\square$

We call the random variable $U_t^\uparrow$ the *time- t superhedging price* of H . Let $\xi \in \Xi$ be a predictable process and B an adapted increasing process with $B_0 = 0$, such that

$$U_t^\uparrow = U_0^\uparrow + \sum_{k=1}^t \xi_k \cdot (X_k - X_{k-1}) - B_t \quad \mathbb{P}\text{-a.s. for } t = 0, \dots, T, \quad (3.12)$$

as in the optional decomposition of $U^\uparrow(H)$. For the final part of this section, we describe the process of consumption B , as in Biagini *et al.* [1, Proposition 4.1]. Define the stochastic process $\tilde{B}$ by

$$\tilde{B}_0 := 0 \quad \text{and} \quad \tilde{B}_t := \text{ess sup } \mathcal{B}_t, \quad t = 1, \dots, T, \quad (3.13)$$

where the set $\mathcal{B}_t$ is defined recursively for $t = 1, \dots, T$ by

$$\mathcal{B}_t := \left\{ D_t \in L^0(\Omega, \mathcal{F}_t, \mathbb{P}) : \tilde{B}_{t-1} \leq D_t \leq U_0^\uparrow + \sum_{k=1}^T \xi_k \cdot \Delta X_k - H \text{ } \mathbb{P}\text{-a.s.} \right\}.$$

We note that $\tilde{B}$ is nonnegative, increasing and adapted by definition. Moreover, notice that by (3.12), the upper bound in the definition of $\mathcal{B}_t$ satisfies

$$U_0^\uparrow + \sum_{k=1}^T \xi_k \cdot \Delta X_k - H = U_T^\uparrow + B_T - H = B_T \geq 0 \quad \mathbb{P}\text{-a.s.}, \quad (3.14)$$

so in particular $\mathcal{B}_t$ is well-defined for all t . Indeed, from the proof of Proposition 3.22 we know that $(U_0^\uparrow, \xi)$ is a superhedging strategy for H with final value

$$V_T = U_0^\uparrow + \sum_{k=1}^T \xi_k \cdot \Delta X_k \geq H \quad \mathbb{P}\text{-a.s.}$$

Theorem 3.24. *Let B be given as in (3.12). Then*

$$B_t = \tilde{B}_t \quad \mathbb{P}\text{-a.s., for } t = 0, \dots, T.$$

Proof. We prove the statement by induction. For $t = 0$, we have $B_0 = 0 = \tilde{B}_0$ by definition. For the induction step, fix $t \in \{1, \dots, T\}$ and assume that

$B_{t-1} = \tilde{B}_{t-1}$ $\mathbb{P}$ -a.s. First, we show that $B_t \leq \tilde{B}_t$ $\mathbb{P}$ -a.s. Note that since B is increasing, we have

$$\tilde{B}_{t-1} = B_{t-1} \leq B_t \leq B_T = U_0^\uparrow + \sum_{k=1}^T \xi_k \cdot \Delta X_k - H \quad \mathbb{P}\text{-a.s.}$$

by the induction hypothesis and by (3.14). Since B is adapted this implies $B_t \in \mathcal{B}_t$, so by definition of the essential supremum we get

$$B_t \leq \tilde{B}_t \quad \mathbb{P}\text{-a.s.} \quad (3.15)$$

We prove the converse inequality. Recall first that by Remark 3.20, the value process V of the self-financing strategy $(U_0^\uparrow, \xi)$, given by

$$V_s = U_0^\uparrow + \sum_{k=1}^s \xi_k \cdot (X_k - X_{k-1}) = U_s^\uparrow + B_s \quad \mathbb{P}\text{-a.s. for } s = 0, \dots, T, \quad (3.16)$$

is a $\mathcal{P}$ -martingale. We define the modified process $\tilde{V}$ by

$$\tilde{V}_s := U_0^\uparrow + \sum_{k=1}^s \xi_k \cdot (X_k - X_{k-1}) - \tilde{B}_s = V_s - \tilde{B}_s, \quad \text{for } s = 0, \dots, T. \quad (3.17)$$

Let $\mathbb{P}^* \in \mathcal{P}$ be arbitrary. Since $\tilde{B}$ is increasing and adapted, for $s = 0, \dots, T-1$ the martingale property of V yields

$$\mathbb{E}^*[\tilde{V}_{s+1} \mid \mathcal{F}_s] = \mathbb{E}^*[V_{s+1} - \tilde{B}_{s+1} \mid \mathcal{F}_s] \leq V_s - \tilde{B}_s = \tilde{V}_s \quad \mathbb{P}\text{-a.s.} \quad (3.18)$$

Furthermore, for $s = 0, \dots, T$ by definition of $\tilde{B}$ we have

$$0 \leq \tilde{B}_s \leq U_0^\uparrow + \sum_{k=1}^T \xi_k \cdot \Delta X_k - H = V_T - H \leq V_T \quad \mathbb{P}\text{-a.s.} \quad (3.19)$$

Since V is a $\mathbb{P}^*$ -martingale, $V_s \in L^1(\Omega, \mathcal{F}_s, \mathbb{P}^*)$ for $s = 0, \dots, T$, and therefore by (3.17) and (3.19) we get $\tilde{V}_s \in L^1(\Omega, \mathcal{F}_s, \mathbb{P}^*)$ for all s . Combined with (3.18), this implies that $\tilde{V}$ is a $\mathbb{P}^*$ -supermartingale. Then, using the supermartingale property and the fact that $\tilde{B}_T \leq V_T - H$ $\mathbb{P}$ -a.s. by (3.19), we obtain

$$\tilde{V}_s \geq \mathbb{E}^*[\tilde{V}_T \mid \mathcal{F}_s] = \mathbb{E}^*[V_T - \tilde{B}_T \mid \mathcal{F}_s] \geq \mathbb{E}^*[H \mid \mathcal{F}_s] \quad \mathbb{P}\text{-a.s., for } s = 0, \dots, T.$$

Since $\mathbb{P}^* \in \mathcal{P}$ was arbitrary, it follows that

$$U_s^\uparrow = \text{ess sup}_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[H \mid \mathcal{F}_s] \leq \tilde{V}_s \quad \mathbb{P}\text{-a.s., for } s = 0, \dots, T. \quad (3.20)$$

Finally, by equations (3.16), (3.17) and (3.20), we conclude

$$\tilde{B}_t = V_t - \tilde{V}_t \leq V_t - U_t^\uparrow = B_t \quad \mathbb{P}\text{-a.s.,}$$

which together with (3.15) shows the induction step and hence concludes the proof. $\square$

For purposes of approximation in Chapter 5, we conclude this section by proving a result reminiscent of Theorem 3.24. Let $\delta > 0$, and for $t = 1, \dots, T$ define the set

$$\mathcal{B}_t^\delta := \left\{ D_t \in L^0(\Omega, \mathcal{F}_t, \mathbb{P}) : -\delta \leq D_t - \delta \leq U_0^\uparrow + \sum_{k=1}^T \xi_k \cdot \Delta X_k - H \text{ } \mathbb{P}\text{-a.s.} \right\}.$$

Lemma 3.25. *Let $\delta > 0$, and let B be given as in (3.12). Then*

$$B_t + \delta = \text{ess sup } \mathcal{B}_t^\delta \quad \mathbb{P}\text{-a.s., for } t = 1, \dots, T.$$

Proof. Fix $t \in \{1, \dots, T\}$. The process B is nonnegative and increasing, so by (3.14) we have

$$-\delta < 0 \leq B_t \leq B_T = U_0^\uparrow + \sum_{k=1}^T \xi_k \cdot \Delta X_k - H \quad \mathbb{P}\text{-a.s.}$$

Since B is adapted it follows that $(B_t + \delta) \in \mathcal{B}_t^\delta$, and thus

$$B_t + \delta \leq \text{ess sup } \mathcal{B}_t^\delta \quad \mathbb{P}\text{-a.s.} \quad (3.21)$$

For the converse inequality, let $\bar{B}_t \in \mathcal{B}_t^\delta$ be arbitrary. We show that $\bar{B}_t \leq B_t + \delta$ $\mathbb{P}$ -a.s. Define $\hat{B}_t := \bar{B}_t \vee \delta$. Then we have $\bar{B}_t \leq \hat{B}_t$ $\mathbb{P}$ -a.s. Moreover, since $\hat{B}_t - \delta = (\bar{B}_t - \delta) \vee 0$ and $\bar{B}_t \in \mathcal{B}_t^\delta$, we obtain

$$0 \leq \hat{B}_t - \delta \leq U_0^\uparrow + \sum_{k=1}^T \xi_k \cdot \Delta X_k - H \quad \mathbb{P}\text{-a.s.}$$

Therefore, we have

$$B_{t-1} \leq (\hat{B}_t - \delta) \vee B_{t-1} \leq U_0^\uparrow + \sum_{k=1}^T \xi_k \cdot \Delta X_k - H \quad \mathbb{P}\text{-a.s.,}$$

so $((\hat{B}_t - \delta) \vee B_{t-1}) \in \mathcal{B}_t$. Hence, we conclude

$$\hat{B}_t - \delta \leq (\hat{B}_t - \delta) \vee B_{t-1} \leq \text{ess sup } \mathcal{B}_t = B_t \quad \mathbb{P}\text{-a.s.,}$$

which implies

$$\bar{B}_t \leq \hat{B}_t \leq B_t + \delta \quad \mathbb{P}\text{-a.s.}$$

Since $\bar{B}_t \in \mathcal{B}_t^\delta$ was arbitrary, this shows

$$\text{ess sup } \mathcal{B}_t^\delta \leq B_t + \delta \quad \mathbb{P}\text{-a.s.,}$$

and together with (3.21) concludes the proof. $\square$

Chapter 4

Quantile hedging

In Chapter 3, we saw that the superhedging price of a European contingent claim H at time 0 is given by the initial value $U_0^\uparrow$ of the upper Snell envelope of H , which coincides with the upper bound of arbitrage-free prices of H . Therefore, following a superhedging strategy eliminates the opportunity for the seller to profit from the claim, together with the risk. Quantile hedging was introduced by Föllmer and Leukert [7] to address this problem. The underlying idea is to reduce the cost of the initial investment by accepting a shortfall probability. More precisely, for a given $\alpha \in (0, 1)$ we aim to minimize the initial cost of a self-financing trading strategy whose terminal value dominates H with probability α , hence accepting a shortfall probability of $1 - \alpha$. In Section 4.1, we present the quantile hedging problem using success sets. Moreover, we show that at time 0, the α -quantile hedging price converges to the superhedging price for α tending to 1. In Section 4.2, we solve an extended optimization problem via random tests and consecutively success ratios.

4.1 Success sets

We begin by introducing quantile hedging via success sets. Our goal is to minimize the initial cost needed to implement a self-financing strategy that is a superhedging strategy for H with probability α , for some fixed $\alpha \in (0, 1)$. First, we provide an equivalent problem to finding the α -quantile hedging price. Then, we use this equivalent formulation to show that for α tending to 1, the α -quantile hedging price converges to the superhedging price $U_0^\uparrow$.

Fix $\alpha \in (0, 1)$, and let H denote a discounted European contingent claim such that

$$\sup_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[H] < \infty,$$

as in the previous chapter. We recall that by Proposition 2.17 and Remark 2.18, each pair $(u, \xi) \in \mathbb{R} \times \Xi$ describes a self-financing trading strategy with value process V given by

$$V_t = u + \sum_{k=1}^t \xi_k \cdot (X_k - X_{k-1}) \quad \mathbb{P}\text{-a.s., for } t = 0, \dots, T.$$

To pose the α -quantile hedging problem, we consider trading strategies whose final value dominates the contingent claim H with probability α . This motivates the following definition.

Definition 4.1. Let $(u, \xi) \in \mathbb{R} \times \Xi$ be a 0-admissible self-financing trading strategy with value process V . Then the set

$$\{\omega \in \Omega : V_T(\omega) \geq H(\omega)\}$$

is called *success set*.

Remark 4.2. We note that as in the definition above, throughout this chapter we restrict ourselves to 0-admissible strategies since this is not automatically given as in the case of superhedging strategies (see also Remark 3.12). Recall that the set of 0-admissible trading strategies is denoted by $\mathcal{A}^0$.

We consider the set $\mathcal{U}_0^\alpha = \mathcal{U}_0^\alpha(H)$, given by

$$\mathcal{U}_0^\alpha := \left\{ u \in \mathbb{R} : \exists \xi \in \Xi \text{ s.t. } (u, \xi) \in \mathcal{A}^0 \text{ and } \mathbb{P} \left[u + \sum_{k=1}^T \xi_k \cdot \Delta X_k \geq H \right] \geq \alpha \right\},$$

representing the initial investments for which there exists a 0-admissible strategy such that the probability of the associated success set is greater than or equal to α . Observe that we have $\mathcal{U}_0(H) \subseteq \mathcal{U}_0^\alpha(H)$ by definition. In particular, $\mathcal{U}_0^\alpha(H)$ is nonempty for each $\alpha \in (0, 1)$, and

$$\inf \mathcal{U}_0^\alpha(H) \leq \inf \mathcal{U}_0(H) = \sup_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[H].$$

We call $\inf \mathcal{U}_0^\alpha(H)$ the α -quantile hedging price of H , and α is called the *probability of success*. We emphasize that the α -quantile hedging problem only takes into account the *shortfall probability* $1 - \alpha$, and not the size of the shortfall. For the latter problem, the work of Föllmer and Leukert [8] gives an outlook.

The following result provides an equivalent to finding the α -quantile hedging price $\inf \mathcal{U}_0^\alpha(H)$. Such an equivalent formulation was first presented in Föllmer and Leukert [7].

Proposition 4.3. Let $\alpha \in (0, 1)$ and let H be a European contingent claim. Then

$$\inf \mathcal{U}_0^\alpha(H) = \inf \left\{ \sup_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[H \mathbb{1}_A] : A \in \mathcal{F}_T, \mathbb{P}(A) \geq \alpha \right\}.$$

Proof. “ $\leq$ ”: Let $A \in \mathcal{F}_T$ be such that $\mathbb{P}[A] \geq \alpha$, and define $\tilde{H} := H \mathbb{1}_A$. Then $\tilde{H}$ is $\mathcal{F}_T$ -measurable since both H and $\mathbb{1}_A$ are $\mathcal{F}_T$ -measurable by definition. Moreover,

$$\sup_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[\tilde{H}] = \sup_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[H \mathbb{1}_A] \leq \sup_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[H] < \infty,$$

so $\tilde{H}$ is a European contingent claim. Hence, by Proposition 3.22, there exists a superhedging strategy $(\tilde{u}, \tilde{\xi})$ for $\tilde{H}$ with $\xi \in \Xi$ and initial investment $\tilde{u} := \sup_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[\tilde{H}]$. Then, the value process $\tilde{V}$ of $(\tilde{u}, \tilde{\xi})$ satisfies

$$\tilde{V}_T = \sup_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[H\mathbb{1}_A] + \sum_{k=1}^T \tilde{\xi}_k \cdot (X_k - X_{k-1}) \geq H\mathbb{1}_A \geq 0 \quad \mathbb{P}\text{-a.s.}$$

In particular, $\tilde{V}_T(\omega) \geq H(\omega)$ for all $\omega \in A$, so we obtain

$$\mathbb{P} \left[\sup_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[H\mathbb{1}_A] + \sum_{k=1}^T \tilde{\xi}_k \cdot (X_k - X_{k-1}) \geq H \right] \geq \mathbb{P}[A] \geq \alpha,$$

which implies $\sup_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[H\mathbb{1}_A] \in \mathcal{U}_0^\alpha(H)$. Since A was arbitrary, we conclude

$$\left\{ \sup_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[H\mathbb{1}_A] : A \in \mathcal{F}_T, \mathbb{P}(A) \geq \alpha \right\} \subseteq \mathcal{U}_0^\alpha(H), \quad (4.1)$$

and thus also

$$\inf \mathcal{U}_0^\alpha(H) \leq \inf \left\{ \sup_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[H\mathbb{1}_A] : A \in \mathcal{F}_T, \mathbb{P}(A) \geq \alpha \right\}.$$

“ $\geq$ ”: Let $\hat{u} \in \mathcal{U}_0^\alpha(H)$ and let $\hat{\xi} \in \Xi$ be the corresponding strategy such that $(\hat{u}, \hat{\xi}) \in \mathcal{A}^0$ and

$$\mathbb{P} \left[\hat{u} + \sum_{k=1}^T \hat{\xi}_k \cdot (X_k - X_{k-1}) \geq H \right] \geq \alpha. \quad (4.2)$$

Define the set

$$\hat{A} := \left\{ \omega \in \Omega : \hat{u} + \sum_{k=1}^T \hat{\xi}_k(\omega) \cdot (X_k(\omega) - X_{k-1}(\omega)) \geq H(\omega) \right\}.$$

Then $\hat{A} \in \mathcal{F}_T$, and $\mathbb{P}[\hat{A}] \geq \alpha$ by (4.2). Moreover, by definition of $\hat{A}$ we have

$$\left(\hat{u} + \sum_{k=1}^T \hat{\xi}_k \cdot (X_k - X_{k-1}) \right) \mathbb{1}_{\hat{A}} \geq H\mathbb{1}_{\hat{A}} \geq 0 \quad \mathbb{P}\text{-a.s.}$$

Since $(\hat{u}, \hat{\xi})$ is 0-admissible, it follows that

$$\left(\hat{u} + \sum_{k=1}^T \hat{\xi}_k \cdot (X_k - X_{k-1}) \right) \mathbb{1}_{\hat{A}^c} \geq 0 \quad \mathbb{P}\text{-a.s.}$$

Therefore, we obtain

$$\hat{u} + \sum_{k=1}^T \hat{\xi}_k \cdot \Delta X_k \geq \left(\hat{u} + \sum_{k=1}^T \hat{\xi}_k \cdot \Delta X_k \right) \mathbb{1}_{\hat{A}} \geq H\mathbb{1}_{\hat{A}} \quad \mathbb{P}\text{-a.s.},$$

which implies $\hat{u} \in \mathcal{U}_0(H\mathbb{1}_{\hat{A}})$. Moreover, $H\mathbb{1}_{\hat{A}}$ defines a European contingent claim so by Proposition 3.22, $\sup_{\mathbb{P}^* \in \mathcal{D}} \mathbb{E}[H\mathbb{1}_{\hat{A}}]$ is the minimal element of $\mathcal{U}_0(H\mathbb{1}_{\hat{A}})$. Thus, we have

$$\hat{u} \geq \sup_{\mathbb{P}^* \in \mathcal{D}} \mathbb{E}[H\mathbb{1}_{\hat{A}}] \geq \inf \left\{ \sup_{\mathbb{P}^* \in \mathcal{D}} \mathbb{E}^*[H\mathbb{1}_A] : A \in \mathcal{F}_T, \mathbb{P}(A) \geq \alpha \right\}.$$

Since $\hat{u} \in \mathcal{U}_0^\alpha(H)$ was arbitrary, we obtain

$$\inf \mathcal{U}_0^\alpha(H) \geq \inf \left\{ \sup_{\mathbb{P}^* \in \mathcal{D}} \mathbb{E}^*[H\mathbb{1}_A] : A \in \mathcal{F}_T, \mathbb{P}(A) \geq \alpha \right\},$$

which concludes the proof. $\square$

For $\alpha \in (0, 1)$, we define the set $\mathcal{F}^\alpha := \{A \in \mathcal{F}_T : \mathbb{P}[A] \geq \alpha\}$. Proposition 4.3 shows that if there exists a suitable set $A^* \in \mathcal{F}^\alpha$ such that

$$\inf_{A \in \mathcal{F}^\alpha} \sup_{\mathbb{P}^* \in \mathcal{D}} \mathbb{E}^*[H\mathbb{1}_A] = \sup_{\mathbb{P}^* \in \mathcal{D}} \mathbb{E}^*[H\mathbb{1}_{A^*}],$$

then by (4.1) we have

$$\inf \mathcal{U}_0^\alpha(H) = \min \mathcal{U}_0^\alpha(H) = \sup_{\mathbb{P}^* \in \mathcal{D}} \mathbb{E}^*[H\mathbb{1}_{A^*}].$$

In particular, the solution to the α -quantile hedging problem would then be given by the solution of the superhedging problem for the modified claim $H\mathbb{1}_{A^*}$. However, Proposition 4.3 does not guarantee the existence of such a set $A^* \in \mathcal{F}^\alpha$. In other words, the α -quantile hedging problem does not always admit a solution.

The following theorem from Biagini *et al.* [1, Theorem 3.4] shows that for α tending to 1, or equivalently the shortfall probability tending to 0, the α -quantile hedging price converges to the superhedging price.

Theorem 4.4. *Let H be a European contingent claim. Then the α -quantile hedging price of H converges to the superhedging price of H as α tends to 1, that is,*

$$\inf \mathcal{U}_0^\alpha(H) \xrightarrow{\alpha \rightarrow 1} \inf \mathcal{U}_0(H).$$

Proof. By Proposition 4.3, we have

$$\inf \mathcal{U}_0^\alpha(H) = \inf_{A \in \mathcal{F}^\alpha} \sup_{\mathbb{P}^* \in \mathcal{D}} \mathbb{E}^*[H\mathbb{1}_A].$$

Moreover, we recall that the superhedging price of H is given by

$$\inf \mathcal{U}_0(H) = U_0^\uparrow = \sup_{\mathbb{P}^* \in \mathcal{D}} \mathbb{E}^*[H].$$

Hence, it suffices to show

$$\inf_{A \in \mathcal{F}^\alpha} \sup_{\mathbb{P}^* \in \mathcal{D}} \mathbb{E}^*[H\mathbb{1}_A] \xrightarrow{\alpha \rightarrow 1} \sup_{\mathbb{P}^* \in \mathcal{D}} \mathbb{E}^*[H]. \quad (4.3)$$

Let $(\alpha_n)_{n \in \mathbb{N}} \subset (0, 1)$ be an increasing sequence with $\lim_{n \rightarrow \infty} \alpha_n = 1$. Then $\mathcal{F}^{\alpha_{n+1}} \subseteq \mathcal{F}^{\alpha_n}$, so we have

$$\inf_{A \in \mathcal{F}^{\alpha_n}} \sup_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[H \mathbb{1}_A] \leq \inf_{A \in \mathcal{F}^{\alpha_{n+1}}} \sup_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[H \mathbb{1}_A] \leq \sup_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[H] < \infty.$$

Thus the sequence $(a_n)_{n \in \mathbb{N}}$ given by $a_n := \inf_{A \in \mathcal{F}^{\alpha_n}} \sup_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[H \mathbb{1}_A]$ is increasing and bounded, so its limit exists and satisfies

$$\lim_{n \rightarrow \infty} a_n \leq \sup_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[H]. \quad (4.4)$$

Fix $\varepsilon > 0$. By definition of the infimum, for each $n \in \mathbb{N}$ there exists $A_n \in \mathcal{F}^{\alpha_n}$ such that

$$\inf_{A \in \mathcal{F}^{\alpha_n}} \sup_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[H \mathbb{1}_A] \leq \sup_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[H \mathbb{1}_{A_n}] < \inf_{A \in \mathcal{F}^{\alpha_n}} \sup_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[H \mathbb{1}_A] + \varepsilon. \quad (4.5)$$

We note that $(\mathbb{1}_{A_n})_{n \in \mathbb{N}} \subset L^\infty(\Omega, \mathcal{F}_T, \mathbb{P}; [0, 1])$ satisfies the assumptions of Lemma 2.8. Thus, there exists a sequence $(\psi_n)_{n \in \mathbb{N}}$, given by

$$\psi_n = \sum_{k=n}^{\infty} \lambda_k^n \mathbb{1}_{A_k} \in \text{conv}(\mathbb{1}_{A_n}, \mathbb{1}_{A_{n+1}}, \dots) \quad (4.6)$$

for some $(\lambda_k^n)_{k=n}^{\infty} \geq 0$ with $\sum_{k=n}^{\infty} \lambda_k^n = 1$ and only finitely many $\lambda_k^n \neq 0$ for each $n \in \mathbb{N}$, and such that (4.6) converges $\mathbb{P}$ -a.s. to some $\psi \in L^\infty(\Omega, \mathcal{F}_T, \mathbb{P}; [0, 1])$ (see also Remark 2.9). By dominated convergence, we obtain

$$\mathbb{E}[\psi] = \lim_{n \rightarrow \infty} \mathbb{E}[\psi_n] = \lim_{n \rightarrow \infty} \mathbb{E} \left[\sum_{k=n}^{\infty} \lambda_k^n \mathbb{1}_{A_k} \right] = \lim_{n \rightarrow \infty} \left(\sum_{k=n}^{\infty} \lambda_k^n \mathbb{E}[\mathbb{1}_{A_k}] \right). \quad (4.7)$$

Recalling that $\sum_{k=n}^{\infty} \lambda_k^n = 1$ for each $n \in \mathbb{N}$, (4.7) yields

$$\begin{aligned} \mathbb{E}[\psi] &\geq \lim_{n \rightarrow \infty} \left(\sum_{k=n}^{\infty} \lambda_k^n \inf_{l \geq n} \mathbb{E}[\mathbb{1}_{A_l}] \right) = \lim_{n \rightarrow \infty} \left(\inf_{l \geq n} \mathbb{E}[\mathbb{1}_{A_l}] \right) \\ &= \liminf_{n \rightarrow \infty} \mathbb{E}[\mathbb{1}_{A_n}] = \liminf_{n \rightarrow \infty} \mathbb{P}[A_n] \geq \liminf_{n \rightarrow \infty} \alpha_n = 1, \end{aligned} \quad (4.8)$$

where the last inequality follows from the fact that $A_n \in \mathcal{F}^{\alpha_n}$ for each $n \in \mathbb{N}$. Since $0 \leq \psi \leq 1$, (4.8) implies $\psi = 1$ $\mathbb{P}$ -a.s. Then, by dominated convergence and using similar arguments for the supremum as for the infimum above, for each $\tilde{\mathbb{P}} \in \mathcal{P}$ we get

$$\begin{aligned} \tilde{\mathbb{E}}[H] &= \tilde{\mathbb{E}}[H\psi] = \lim_{n \rightarrow \infty} \tilde{\mathbb{E}}[H\psi_n] = \lim_{n \rightarrow \infty} \tilde{\mathbb{E}} \left[H \sum_{k=n}^{\infty} \lambda_k^n \mathbb{1}_{A_k} \right] \\ &= \lim_{n \rightarrow \infty} \left(\sum_{k=n}^{\infty} \lambda_k^n \tilde{\mathbb{E}}[H \mathbb{1}_{A_k}] \right) \leq \lim_{n \rightarrow \infty} \left(\sum_{k=n}^{\infty} \lambda_k^n \sup_{l \geq n} \tilde{\mathbb{E}}[H \mathbb{1}_{A_l}] \right) \\ &= \limsup_{n \rightarrow \infty} \tilde{\mathbb{E}}[H \mathbb{1}_{A_n}] \leq \limsup_{n \rightarrow \infty} \sup_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[H \mathbb{1}_{A_n}] \\ &< \limsup_{n \rightarrow \infty} \left(\inf_{A \in \mathcal{F}^{\alpha_n}} \sup_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[H \mathbb{1}_A] + \varepsilon \right) = \lim_{n \rightarrow \infty} a_n + \varepsilon, \end{aligned} \quad (4.9)$$

where the penultimate step follows from (4.5). Since $\lim_{n \rightarrow \infty} a_n$ exists and (4.9) holds for any $\tilde{\mathbb{P}} \in \mathcal{P}$, we obtain

$$\sup_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[H] \leq \lim_{n \rightarrow \infty} a_n + \varepsilon. \quad (4.10)$$

Since $\varepsilon > 0$ was arbitrary, (4.4) and (4.10) yield

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \left(\inf_{A \in \mathcal{F}^{\alpha_n}} \sup_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[H \mathbb{1}_A] \right) = \sup_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[H],$$

which shows (4.3) and thus concludes the proof. $\square$

4.2 Success ratios

In the previous section, for a European contingent claim H and $\alpha \in (0, 1)$ we introduced the α -quantile hedging problem, and saw it is equivalent to the superhedging problem of the modified contingent claim $H \mathbb{1}_{A^*}$, for some suitable set $A^* \in \mathcal{F}^\alpha$. However, it is a priori not clear whether such a set exists. In this section, we extend our minimization problem to a larger set of $\mathcal{F}_T$ measurable random variables on $[0, 1]$, namely randomized tests. Furthermore, we introduce success ratios and prove that they solve our extended problem. For Propositions 4.5 and 4.8 below, we follow Föllmer and Leukert [7] and Biagini *et al.* [1].

Let $\alpha \in (0, 1)$ and let H denote a European contingent claim. We recall that by Proposition 4.3, we have the equivalent formulation of the α -quantile hedging price of H

$$\inf \mathcal{U}_0^\alpha(H) = \inf_{A \in \mathcal{F}^\alpha} \sup_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[H \mathbb{1}_A]. \quad (4.11)$$

In the following, we extend the minimization problem on the right-hand-side of (4.11). Denote by $\mathcal{R} := L^\infty(\Omega, \mathcal{F}_T, \mathbb{P}; [0, 1])$ the set of randomized tests. For $\alpha \in (0, 1)$, we define

$$\mathcal{R}^\alpha := \{\varphi \in \mathcal{R} : \mathbb{E}[\varphi] \geq \alpha\},$$

and consider the minimization problem

$$\inf \left\{ \sup_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[H \varphi] : \varphi \in \mathcal{R}^\alpha \right\}. \quad (4.12)$$

Note that we have $\{\mathbb{1}_A : A \in \mathcal{F}^\alpha\} \subseteq \mathcal{R}^\alpha$, implying

$$\inf_{\varphi \in \mathcal{R}^\alpha} \sup_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[H \varphi] \leq \inf_{A \in \mathcal{F}^\alpha} \sup_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[H \mathbb{1}_A].$$

We now prove that the extended problem (4.12) admits a solution.

Proposition 4.5. *The infimum in the minimization problem (4.12) is attained, i.e., there exists a randomized test $\tilde{\varphi} \in \mathcal{R}^\alpha \subset \mathcal{R}$ such that*

$$\mathbb{E}[\tilde{\varphi}] = \alpha,$$

and

$$\inf_{\varphi \in \mathcal{R}^\alpha} \sup_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[H\varphi] = \sup_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[H\tilde{\varphi}]. \quad (4.13)$$

Proof. Let $(\varphi_n)_{n \in \mathbb{N}} \subset \mathcal{R}^\alpha$ be a minimizing sequence, i.e., a sequence satisfying

$$\lim_{n \rightarrow \infty} \sup_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[H\varphi_n] = \inf_{\varphi \in \mathcal{R}^\alpha} \sup_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[H\varphi]. \quad (4.14)$$

By Lemma 2.8, there exists a sequence of convex combinations $(\tilde{\varphi}_n)_{n \in \mathbb{N}}$, given by

$$\tilde{\varphi}_n = \sum_{k=n}^{\infty} \lambda_k^n \varphi_k \in \text{conv}(\varphi_n, \varphi_{n+1}, \dots) \quad (4.15)$$

for some $(\lambda_k^n)_{k=n}^{\infty} \geq 0$ with $\sum_{k=n}^{\infty} \lambda_k^n = 1$ and only finitely many $\lambda_k^n \neq 0$ for each $n \in \mathbb{N}$, and such that (4.15) converges $\mathbb{P}$ -a.s. to some $\tilde{\varphi} \in \mathcal{R}$. Since $\varphi_n \in \mathcal{R}^\alpha$ for each $n \in \mathbb{N}$, we have

$$\mathbb{E}[\tilde{\varphi}_n] = \mathbb{E}\left[\sum_{k=n}^{\infty} \lambda_k^n \varphi_k\right] = \sum_{k=n}^{\infty} \lambda_k^n \mathbb{E}[\varphi_k] \geq \sum_{k=n}^{\infty} \lambda_k^n \inf_{l \geq n} \mathbb{E}[\varphi_l] = \inf_{l \geq n} \mathbb{E}[\varphi_l] \geq \alpha,$$

so $\tilde{\varphi}_n \in \mathcal{R}^\alpha$, for each $n \in \mathbb{N}$. Hence, using dominated convergence we get

$$\mathbb{E}[\tilde{\varphi}] = \lim_{n \rightarrow \infty} \mathbb{E}[\tilde{\varphi}_n] \geq \alpha, \quad (4.16)$$

which implies $\tilde{\varphi} \in \mathcal{R}^\alpha$. In particular, it follows that

$$\inf_{\varphi \in \mathcal{R}^\alpha} \sup_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[H\varphi] \leq \sup_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[H\tilde{\varphi}]. \quad (4.17)$$

We show the converse inequality to get (4.13). Let $\tilde{\mathbb{P}} \in \mathcal{P}$ be arbitrary. Again by dominated convergence, we have

$$\begin{aligned} \tilde{\mathbb{E}}[H\tilde{\varphi}] &= \lim_{n \rightarrow \infty} \tilde{\mathbb{E}}[H\tilde{\varphi}_n] = \lim_{n \rightarrow \infty} \tilde{\mathbb{E}}\left[H \sum_{k=n}^{\infty} \lambda_k^n \varphi_k\right] \\ &= \lim_{n \rightarrow \infty} \left(\sum_{k=n}^{\infty} \lambda_k^n \tilde{\mathbb{E}}[H\varphi_k]\right) \leq \lim_{n \rightarrow \infty} \left(\sup_{l \geq n} \tilde{\mathbb{E}}[H\varphi_l]\right) \\ &= \limsup_{n \rightarrow \infty} \tilde{\mathbb{E}}[H\varphi_n]. \end{aligned} \quad (4.18)$$

Hence, by (4.14) we get

$$\begin{aligned} \inf_{\varphi \in \mathcal{R}^\alpha} \sup_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[H\varphi] &= \lim_{n \rightarrow \infty} \sup_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[H\varphi_n] = \limsup_{n \rightarrow \infty} \sup_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[H\varphi_n] \\ &\geq \limsup_{n \rightarrow \infty} \tilde{\mathbb{E}}[H\varphi_n] \geq \tilde{\mathbb{E}}[H\tilde{\varphi}], \end{aligned} \quad (4.19)$$

where in the last inequality we used (4.18). Since $\tilde{\mathbb{P}} \in \mathcal{P}$ was arbitrary, (4.19) implies

$$\inf_{\varphi \in \mathcal{R}^\alpha} \sup_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[H\varphi] \geq \sup_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[H\tilde{\varphi}],$$

which combined with (4.17) proves

$$\inf_{\varphi \in \mathcal{R}^\alpha} \sup_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[H\varphi] = \sup_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[H\tilde{\varphi}].$$

It remains to show that $\mathbb{E}[\tilde{\varphi}] = \alpha$. Recall that $\mathbb{E}[\tilde{\varphi}] \geq \alpha$ by (4.16). Suppose for contradiction that $\mathbb{E}[\tilde{\varphi}] > \alpha$. Then there exists $\varepsilon > 0$ such that $\mathbb{E}[\tilde{\varphi}](1 - \varepsilon) \geq \alpha$, and hence $\varphi_\varepsilon := \tilde{\varphi}(1 - \varepsilon) \in \mathcal{R}^\alpha$. Therefore,

$$\begin{aligned} \sup_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[H\varphi_\varepsilon] &= \sup_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[H\tilde{\varphi}(1 - \varepsilon)] = (1 - \varepsilon) \sup_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[H\tilde{\varphi}] \\ &< \sup_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[H\tilde{\varphi}] = \inf_{\varphi \in \mathcal{R}^\alpha} \sup_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[H\varphi], \end{aligned}$$

yielding a contradiction to $\varphi_\varepsilon \in \mathcal{R}^\alpha$. Thus it follows that $\mathbb{E}[\tilde{\varphi}] = \alpha$, which concludes the proof. $\square$

We now present a special form of randomized tests and reformulate the minimization problem (4.12) accordingly. Denote the set of value processes of all 0-admissible trading strategies by

$$\mathcal{V} := \left\{ V = (V_t)_{t=0, \dots, T} : \exists \bar{\xi} \in \mathcal{A}^0 \text{ s.t. } V \text{ is the value process of } \bar{\xi} \right\}.$$

The motivation behind the following definition comes from the Neyman-Pearson lemma, a fundamental result of statistical testing. For a more complete view, we refer the reader to Föllmer and Leukert [7].

Definition 4.6. The *success ratio* of a 0-admissible trading strategy with value process $V \in \mathcal{V}$ is defined as

$$\varphi_V := \mathbb{1}_{\{V_T \geq H\}} + \frac{V_T}{H} \mathbb{1}_{\{V_T < H\}}. \quad (4.20)$$

Remark 4.7. We note that $\mathbb{P}[\{H = 0\} \cap \{V_T < H\}] = 0$, since $V_T \geq 0$ $\mathbb{P}$ -a.s. for all $V \in \mathcal{V}$. Hence, (4.20) is well-defined.

We have $\varphi_V \in \mathcal{R}$ for all $V \in \mathcal{V}$ by definition. For $\alpha \in (0, 1)$, we define

$$\mathcal{V}^\alpha := \{V \in \mathcal{V} : \mathbb{E}[\varphi_V] \geq \alpha\}.$$

Then we note that $\{\varphi_V \in \mathcal{R} : V \in \mathcal{V}^\alpha\} \subseteq \mathcal{R}^\alpha$. We consider the minimization problem

$$\inf \left\{ \sup_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[H\varphi_V] : V \in \mathcal{V}^\alpha \right\}. \quad (4.21)$$

Let $\tilde{\varphi} \in \mathcal{R}^\alpha$ be as in Proposition 4.5, i.e., such that

$$\inf_{\varphi \in \mathcal{R}^\alpha} \sup_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[H\varphi] = \sup_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[H\tilde{\varphi}] \quad \text{and} \quad \mathbb{E}[\tilde{\varphi}] = \alpha.$$

We show that (4.21) is equivalent to (4.12), or in other words, $\tilde{\varphi}$ is the success ratio associated to some trading strategy with value process $V \in \mathcal{V}^\alpha$.

Proposition 4.8. *There exists a 0-admissible self-financing trading strategy with value process $\tilde{V} \in \mathcal{V}^\alpha$ such that the associated success ratio $\varphi_{\tilde{V}}$ satisfies*

$$\varphi_{\tilde{V}} = \tilde{\varphi} \quad \mathbb{P}\text{-a.s.}$$

Furthermore, $\varphi_{\tilde{V}}$ solves (4.21), i.e., $\mathbb{E}[\varphi_{\tilde{V}}] = \alpha$ and

$$\inf_{V \in \mathcal{V}^\alpha} \sup_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[H\varphi_V] = \sup_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[H\varphi_{\tilde{V}}]. \quad (4.22)$$

Proof. Let $\tilde{H} := H\tilde{\varphi}$. Then, $\tilde{H}$ is also a European contingent claim as in Definition 3.1. Hence, by Proposition 3.22 there exists a minimal (0-admissible) superhedging strategy $\tilde{\xi}$ for $\tilde{H}$, with value process $\tilde{V} \in \mathcal{V}$ and initial investment $\tilde{V}_0 = \sup_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[\tilde{H}]$. Moreover, using that $\tilde{V}_T \geq \tilde{H} = H\tilde{\varphi}$ $\mathbb{P}$ -a.s. and $\tilde{\varphi} \leq 1$, we have

$$\begin{aligned} \varphi_{\tilde{V}} &= \mathbb{1}_{\{\tilde{V}_T \geq H\}} + \frac{\tilde{V}_T}{H} \mathbb{1}_{\{\tilde{V}_T < H\}} \\ &\geq \tilde{\varphi} \mathbb{1}_{\{\tilde{V}_T \geq H\}} + \frac{H\tilde{\varphi}}{H} \mathbb{1}_{\{\tilde{V}_T < H\}} = \tilde{\varphi} \quad \mathbb{P}\text{-a.s.} \end{aligned} \quad (4.23)$$

In particular, since $\tilde{\varphi} \in \mathcal{R}^\alpha$ taking expectations yields

$$\mathbb{E}[\varphi_{\tilde{V}}] \geq \mathbb{E}[\tilde{\varphi}] \geq \alpha, \quad (4.24)$$

and thus $\tilde{V} \in \mathcal{V}^\alpha$. We show that $\varphi_{\tilde{V}} \leq \tilde{\varphi}$ $\mathbb{P}$ -a.s. From Theorem 2.21 we know that the value process $\tilde{V}$ is a $\mathcal{P}$ -martingale. It follows that $\mathbb{E}^*[\tilde{V}_T] = \tilde{V}_0$ for each $\mathbb{P}^* \in \mathcal{P}$, and therefore

$$\begin{aligned} \sup_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[H\varphi_{\tilde{V}}] &= \sup_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*\left[H\left(\mathbb{1}_{\{\tilde{V}_T \geq H\}} + \frac{\tilde{V}_T}{H} \mathbb{1}_{\{\tilde{V}_T < H\}}\right)\right] \\ &\leq \sup_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[\tilde{V}_T] \\ &= \tilde{V}_0 \\ &= \sup_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[H\tilde{\varphi}] \\ &= \inf_{\varphi \in \mathcal{R}^\alpha} \sup_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[H\varphi]. \end{aligned} \quad (4.25)$$

Since $\varphi_{\tilde{V}} \in \mathcal{R}^\alpha$ by (4.24), we have

$$\inf_{\varphi \in \mathcal{R}^\alpha} \sup_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[H\varphi] \leq \sup_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[H\varphi_{\tilde{V}}],$$

which combined with (4.25) implies that $\varphi_{\tilde{V}}$ is another minimizer, i.e.,

$$\inf_{\varphi \in \mathcal{R}^\alpha} \sup_{\mathbb{P}^* \in \mathcal{D}} \mathbb{E}^*[H\varphi] = \sup_{\mathbb{P}^* \in \mathcal{D}} \mathbb{E}^*[H\varphi_{\tilde{V}}].$$

Then, by the same argument as in the proof of Proposition 4.5, we must have

$$\mathbb{E}[\varphi_{\tilde{V}}] = \alpha,$$

otherwise we could find $\varepsilon > 0$ such that $\varphi_{\tilde{V}}(1 - \varepsilon) \in \mathcal{R}^\alpha$ and

$$\sup_{\mathbb{P}^* \in \mathcal{D}} \mathbb{E}^*[H\varphi_{\tilde{V}}(1 - \varepsilon)] < \sup_{\mathbb{P}^* \in \mathcal{D}} \mathbb{E}^*[H\varphi_{\tilde{V}}] = \inf_{\varphi \in \mathcal{R}^\alpha} \sup_{\mathbb{P}^* \in \mathcal{D}} \mathbb{E}^*[H\varphi],$$

yielding a contradiction. Hence we obtain

$$\mathbb{E}[\varphi_{\tilde{V}} - \tilde{\varphi}] = \alpha - \alpha = 0,$$

which together with (4.23) shows that $\varphi_{\tilde{V}} = \tilde{\varphi}$ $\mathbb{P}$ -a.s. It remains to show (4.22).

Since $\{\varphi_V \in \mathcal{R} : V \in \mathcal{V}^\alpha\} \subseteq \mathcal{R}^\alpha$, we have

$$\sup_{\mathbb{P}^* \in \mathcal{D}} \mathbb{E}^*[H\varphi_{\tilde{V}}] = \inf_{\varphi \in \mathcal{R}^\alpha} \sup_{\mathbb{P}^* \in \mathcal{D}} \mathbb{E}^*[H\varphi] \leq \inf_{V \in \mathcal{V}^\alpha} \sup_{\mathbb{P}^* \in \mathcal{D}} \mathbb{E}^*[H\varphi_V].$$

Conversely, since $\tilde{V} \in \mathcal{V}^\alpha$ we get

$$\inf_{V \in \mathcal{V}^\alpha} \sup_{\mathbb{P}^* \in \mathcal{D}} \mathbb{E}^*[H\varphi_V] \leq \sup_{\mathbb{P}^* \in \mathcal{D}} \mathbb{E}^*[H\varphi_{\tilde{V}}],$$

which shows (4.22) and hence concludes the proof. $\square$

We have thus proved that the optimization problems (4.12) and (4.21) are equivalent and admit a solution $\varphi_{\tilde{V}}$, that is a success ratio associated to some 0-admissible trading strategy with value process $\tilde{V} \in \mathcal{V}^\alpha$. Since $\{\mathbb{1}_A : A \in \mathcal{F}^\alpha\} \subseteq \mathcal{R}^\alpha$, using the equivalent formulation of the α -quantile hedging price from Proposition 4.3 we have

$$\begin{aligned} \sup_{\mathbb{P}^* \in \mathcal{D}} \mathbb{E}^*[H\varphi_{\tilde{V}}] &= \inf_{V \in \mathcal{V}^\alpha} \sup_{\mathbb{P}^* \in \mathcal{D}} \mathbb{E}^*[H\varphi_V] = \inf_{\varphi \in \mathcal{R}^\alpha} \sup_{\mathbb{P}^* \in \mathcal{D}} \mathbb{E}^*[H\varphi] \\ &\leq \inf_{A \in \mathcal{F}^\alpha} \sup_{\mathbb{P}^* \in \mathcal{D}} \mathbb{E}^*[H\mathbb{1}_A] = \inf \mathcal{U}_0^\alpha(H). \end{aligned}$$

Hence, we obtain the bounds

$$\sup_{\mathbb{P}^* \in \mathcal{D}} \mathbb{E}^*[H\varphi_{\tilde{V}}] \leq \inf \mathcal{U}_0^\alpha(H) \leq \sup_{\mathbb{P}^* \in \mathcal{D}} \mathbb{E}^*[H]$$

for the α -quantile hedging price.

To conclude this chapter, we prove that for α tending to 1, the minimization problem (4.21) converges to the superhedging price. The proof of Proposition 4.9 is similar to the proof of Theorem 4.4, so we give it in the Appendix.

Proposition 4.9. *Let φ_V denote the success ratio associated to a 0-admissible strategy $\tilde{\xi}$ with value process V . Then*

$$\inf_{V \in \mathcal{V}^\alpha} \sup_{\mathbb{P}^* \in \mathcal{D}} \mathbb{E}^*[H\varphi_V] \xrightarrow{\alpha \rightarrow 1} \sup_{\mathbb{P}^* \in \mathcal{D}} \mathbb{E}^*[H].$$

Chapter 5

Neural network approximations

The goal of this chapter is to approximate the superhedging price process of a discounted European contingent claim H by neural networks, using the results obtained in Chapters 3 and 4. In Section 5.1, we present the definition and approximation properties of multilayer feedforward neural networks. Section 5.2 contains the main approximation result of this chapter, namely that under suitable assumptions the superhedging price can be approximated by neural networks. Finally, Section 5.3 focuses on the approximation of the process of consumption B from the optional decomposition of the superhedging price process of the claim H . The approximation results contained in this chapter follow Biagini *et al.* [1], where the interested reader can also find the respective numerical results.

5.1 Introduction to feedforward neural networks

This section introduces the mathematical model of a multilayer feedforward neural network. We highlight the universal approximation properties of such a model and recall the approximation results relevant for the following sections, in particular for L^p functions.

Neural networks were originally introduced by McCulloch and Pitts [15] in 1943 and are loosely inspired by the human brain. In broad terms, a neural network consists of neurons arranged in layers and connected by weighted edges. Mathematically this is represented by the composition of affine functions with a nonlinear element. The number of layers of a neural network defines the depth of the model. Typically, these layers are vector valued, and each element of the vector resembles the role of a neuron, receiving inputs from many other units (from the previous layer) and computing its own output. Feedforward neural networks are characterized by the fact that they do not have feedback connections in which the outputs of the model are returned into itself. For a further outlook on neural networks we refer the reader to the book of Goodfellow *et al.* [10].

Today deep neural networks represent the foundation of machine learning and artificial intelligence. Due to the important progress made in recent years in

this area, neural network models have gained importance also in mathematical finance. Ruf and Wang [16] offers an overview of neural network applications to option pricing and hedging. For the definition of a multilayer feedforward neural network, we follow Buehler *et al.* [2].

Definition 5.1. Let $L, d_0, d_1, \dots, d_L \in \mathbb{N}$ with $L \geq 2$. For $l = 1, \dots, L$, let $W_l : \mathbb{R}^{d_{l-1}} \rightarrow \mathbb{R}^{d_l}$ be an affine function, given by $W_l = A^l x + b^l$ for some $A^l \in \mathbb{R}^{d_l \times d_{l-1}}$ and $b^l \in \mathbb{R}^{d_l}$. Let $\sigma : \mathbb{R} \rightarrow \mathbb{R}$ be a (Borel-)measurable function, and for $l = 1, \dots, L-1$ define $F_l := \sigma \circ W_l$, with σ applied componentwise. Then a function $F : \mathbb{R}^{d_0} \rightarrow \mathbb{R}^{d_L}$ defined as

$$F = W_L \circ F_{L-1} \circ \dots \circ F_1$$

is called *feedforward neural network*.

In Definition 5.1, L denotes the number of layers, $d_1, \dots, d_{L-1}$ denote the dimensions of the hidden layers, and d_0, d_L are the dimension of the input and output layers, respectively. For any $i = 1, \dots, d_l, j = 1, \dots, d_{l-1}$, the number A_{ij}^l is interpreted as the weight of the edge connecting the node i of layer $l-1$ to node j of layer l . Moreover, the function σ is called *activation function*. Common choices for σ include $\sigma(x) := \frac{1}{1+e^{-x}}$, $\sigma(x) := \tanh(x)$, and $\sigma(x) := \max(0, x)$. The purpose of the activation function is to introduce nonlinearity into the model. We denote the set of neural networks mapping $\mathbb{R}^{d_i} \rightarrow \mathbb{R}^{d_o}$ with activation function σ by

$$\mathcal{NN}_{d_i, d_o}^\sigma,$$

for some $d_i, d_o \in \mathbb{N}$, representing the dimensions of the input and output layers, respectively.

The strong approximation properties of neural networks have been widely studied in the literature. The following result, Hornik [11, Theorem 1], shows that the set of neural networks with one hidden layer and one output unit is dense in L^p . For the more general case of multilayer feedforward neural networks with d_o output units, the same result can be deduced from Hornik *et al.* [12, Corollaries 2.6 and 2.7]. Corollary 5.3 below states the precise approximation result we use throughout the remaining part of this chapter, and we present its proof in the Appendix.

Theorem 5.2 (Universal approximation). *Suppose the activation function σ is bounded and nonconstant. Then for any finite measure μ on $(\mathbb{R}^{d_i}, \mathcal{B}(\mathbb{R}^{d_i}))$ and for $1 \leq p < \infty$, the set $\mathcal{NN}_{d_i, 1}^\sigma$ is dense in $L^p(\mathbb{R}^{d_i}, \mathcal{B}(\mathbb{R}^{d_i}), \mu)$.*

Corollary 5.3. *Suppose the activation function σ is bounded and nonconstant. Let $f : (\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d)) \rightarrow (\mathbb{R}^m, \mathcal{B}(\mathbb{R}^m))$, $d, m \in \mathbb{N}$, be a measurable function and let μ be a probability measure on $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$. Then for any $\delta, \varepsilon > 0$ there exists a neural network $g \in \mathcal{NN}_{d, m}^\sigma$ such that*

$$\mu[\{x \in \mathbb{R}^d : \|f(x) - g(x)\| > \delta\}] < \varepsilon.$$

5.2 Approximation of the superhedging price at time $t = 0$

In this section, we prove that the superhedging price of a European contingent claim H at time 0 can be approximated by feedforward neural networks. To do this, we first show that for a suitable filtration we can approximate adapted and predictable processes by neural networks. Using this, we construct the approximate superhedging and α -quantile hedging price of the claim H for $\alpha \in (0, 1)$, which then allows us to show the desired approximation of the superhedging price.

From now on, we assume that the filtration $\mathbb{F} = (\mathcal{F}_t)_{t=0,\dots,T}$ is given by $\mathcal{F}_t = \sigma(Y_0, \dots, Y_t)$, for $t = 0, \dots, T$, where $Y = (Y_k)_{k=0,\dots,T}$ is some $\mathbb{R}^m$ -valued process, for some $m \in \mathbb{N}$, such that Y_0 is a constant and $\sigma(Y_0, \dots, Y_T) = \mathcal{F}$. Note that with this definition of the filtration, $\mathcal{F}_0$ is trivial and $\mathcal{F}_T = \mathcal{F}$, and thus $\mathbb{F}$ satisfies the assumptions on the general filtration in our model from the previous chapters. To simplify notation, we write $\mathbf{Y}_t := (Y_0, \dots, Y_t)$ for $t = 0, \dots, T$. Furthermore, we recall that by Lemma 2.12, for $t = 0, \dots, T$, any $\mathcal{F}_t$ -measurable random variable Z can be written as

$$Z = f_t(\mathbf{Y}_t) = f_t(Y_0, \dots, Y_t), \quad (5.1)$$

for some (Borel-)measurable function $f_t : \mathbb{R}^{m(t+1)} \rightarrow \mathbb{R}^d$. Denote by

$$\Theta_t := \bigcup_{L \geq 2} \left(\bigcup_{(N_0, \dots, N_L) \in \{m(t+1)\} \times \mathbb{N}^{L-1} \times \{d\}} \left(\times_{l=1}^L (\mathbb{R}^{N_l \times N_{l-1}} \times \mathbb{R}^{N_l}) \right) \right)$$

the set of network parameters corresponding to neural networks mapping $\mathbb{R}^{m(t+1)} \rightarrow \mathbb{R}^d$, for $t = 0, \dots, T$. Moreover, F^{θ_t} denotes the neural network with parameters $\theta_t \in \Theta_t$, or in other words,

$$\mathcal{NN}_{m(t+1),d}^\sigma = \{F^{\theta_t} : \theta_t \in \Theta_t\}.$$

Let $\alpha \in (0, 1)$ and let H denote a European contingent claim. We recall that the time-0 superhedging and α -quantile hedging prices of H are given by

$$\inf \mathcal{U}_0 := \inf \left\{ u \in \mathbb{R} : \exists \xi \in \Xi \text{ s.t. } u + \sum_{k=1}^T \xi_k \cdot \Delta X_k \geq H \text{ } \mathbb{P}\text{-a.s.} \right\}$$

and

$$\inf \mathcal{U}_0^\alpha := \inf \left\{ u \in \mathbb{R} : \exists \xi \in \Xi \text{ s.t. } (u, \xi) \in \mathcal{A}^0, \mathbb{P} \left[u + \sum_{k=1}^T \xi_k \cdot \Delta X_k \geq H \right] \geq \alpha \right\},$$

respectively. Since each $\xi = (\xi_t)_{t=1,\dots,T} \in \Xi$ is $\mathbb{F}$ -predictable, by (5.1) for every $t = 1, \dots, T$ there exists a measurable function $f_{t-1} : \mathbb{R}^{mt} \rightarrow \mathbb{R}^d$ such that

$$\xi_t = f_{t-1}(\mathbf{Y}_{t-1}) = f_{t-1}(Y_0, \dots, Y_{t-1}).$$

5.2. Approximation of the superhedging price at time $t = 0$

For $t = 1, \dots, T$, let μ_t be the probability measure given by the distribution of $\mathbf{Y}_{t-1}$ under $\mathbb{P}$, i.e.,

$$\mu_t(B) = \mathbb{P}[\{\omega \in \Omega : \mathbf{Y}_{t-1}(\omega) \in B\}],$$

for $B \in \mathcal{B}(\mathbb{R}^{mt})$. Then, by Corollary 5.3, for any $\varepsilon, \delta > 0$ there exists some parameter $\theta_{t-1, \varepsilon} \in \Theta_{t-1}$ such that the corresponding neural network $F^{\theta_{t-1, \varepsilon}}(\mathbf{Y}_{t-1}) \in \mathcal{NN}_{mt, d}^\sigma$ satisfies

$$\begin{aligned} \varepsilon &> \mu_t[\{x \in \mathbb{R}^{mt} : \|f_{t-1}(x) - F^{\theta_{t-1, \varepsilon}}(x)\| > \delta\}] \\ &= \mathbb{P}[\{\omega \in \Omega : \|f_{t-1}(\mathbf{Y}_{t-1}(\omega)) - F^{\theta_{t-1, \varepsilon}}(\mathbf{Y}_{t-1}(\omega))\| > \delta\}]. \end{aligned}$$

For $\varepsilon, \delta \rightarrow 0$, this implies that ξ_t can be approximated arbitrarily well in $L^0(\Omega, \mathcal{F}_{t-1}, \mathbb{P})$ by $F^{\theta_{t-1, \varepsilon}}(\mathbf{Y}_{t-1})$, for $t = 1, \dots, T$. This motivates the following definition.

Definition 5.4. For $\alpha \in (0, 1)$, we define the sets

$$\begin{aligned} \mathcal{U}_0^\Theta(H) := & \left\{ u \in \mathbb{R} : \exists \theta_{k-1, \varepsilon} \in \Theta_{k-1}, k = 1, \dots, T, \text{ s.t.} \right. \\ & \left. u + \sum_{k=1}^T F^{\theta_{k-1, \varepsilon}}(\mathbf{Y}_{k-1}) \cdot \Delta X_k \geq H \text{ } \mathbb{P}\text{-a.s.} \right\} \end{aligned}$$

and

$$\begin{aligned} \mathcal{U}_0^{\Theta, \alpha}(H) := & \left\{ u \in \mathbb{R} : \exists \theta_{k-1, \varepsilon} \in \Theta_{k-1}, k = 1, \dots, T, \text{ s.t.} \right. \\ & \left. \mathbb{P} \left[u + \sum_{k=1}^T F^{\theta_{k-1, \varepsilon}}(\mathbf{Y}_{k-1}) \cdot \Delta X_k \geq H \right] \geq \alpha \right\}. \end{aligned}$$

Then we call $\inf \mathcal{U}_0^\Theta(H)$ the *approximate superhedging price* of H , and $\inf \mathcal{U}_0^{\Theta, \alpha}(H)$ the *approximate α -quantile hedging price* of H .

Note that by the results obtained in the paragraph above, we have

$$\mathcal{U}_0(H) \subseteq \mathcal{U}_0^\Theta(H) \quad \text{and} \quad \mathcal{U}_0^\alpha(H) \subseteq \mathcal{U}_0^{\Theta, \alpha}(H).$$

In particular, $\mathcal{U}_0^\Theta(H)$ and $\mathcal{U}_0^{\Theta, \alpha}(H)$ are nonempty. For $C > 0$ and for $\theta_t \in \Theta_t$ with corresponding neural network F^{θ_t} , we define the *truncated neural network*

$$F^{\theta_t, C} := (F^{\theta_t} \wedge C) \vee (-C), \tag{5.2}$$

and the sets

$$\begin{aligned} \mathcal{U}_0^{\Theta, C}(H) := & \left\{ u \in \mathbb{R} : \exists \theta_{k-1, \varepsilon} \in \Theta_{k-1}, k = 1, \dots, T, \text{ s.t.} \right. \\ & \left. u + \sum_{k=1}^T F^{\theta_{k-1, \varepsilon}, C}(\mathbf{Y}_{k-1}) \cdot \Delta X_k \geq H \text{ } \mathbb{P}\text{-a.s.} \right\} \end{aligned}$$

and

$$\mathcal{U}_0^{\Theta, C, \alpha}(H) := \left\{ u \in \mathbb{R} : \exists \theta_{k-1, \xi} \in \Theta_{k-1}, k = 1, \dots, T, \text{ s.t. } \right. \\ \left. \mathbb{P} \left[u + \sum_{k=1}^T F^{\theta_{k-1, \xi}, C}(\mathbf{Y}_{\mathbf{k}-1}) \cdot \Delta X_k \geq H \right] \geq \alpha \right\}.$$

Then the *truncated approximate superhedging and α -quantile hedging prices* of H are given by

$$\inf \mathcal{U}_0^{\Theta, C}(H) \quad \text{and} \quad \inf \mathcal{U}_0^{\Theta, C, \alpha}(H),$$

respectively.

We note that one cannot make sure that the neural network approximations of the trading strategies are admissible. For this reason, we introduce the *superhedging price of H restricted to bounded strategies*,

$$\inf \mathcal{U}_0^{bdd}(H) := \inf \left\{ u \in \mathbb{R} : \exists \xi \in \Xi \text{ s.t. } \xi_k \in L^\infty(\Omega, \mathcal{F}_{k-1}, \mathbb{P}), k = 1, \dots, T, \right. \\ \left. \text{and } u + \sum_{k=1}^T \xi_k \cdot (X_k - X_{k-1}) \geq H \text{ } \mathbb{P}\text{-a.s.} \right\}.$$

Under the following assumption, the superhedging price of H does not change if we restrict it to bounded strategies.

Assumption 5.5. The superhedging price of the contingent claim H coincides with its superhedging price restricted to bounded strategies, that is,

$$\inf \mathcal{U}_0(H) = \inf \mathcal{U}_0^{bdd}(H).$$

Remark 5.6. Note that in particular Assumption 5.5 implies that $\mathcal{U}_0^{bdd}(H) \neq \emptyset$, with the convention $\sup \emptyset = -\infty$, $\inf \emptyset = \infty$. We recall that

$$\sup_{\mathbb{P}^* \in \mathcal{D}} \mathbb{E}^*[H] = \min \mathcal{U}_0(H) = \inf \mathcal{U}_0(H) = \inf \mathcal{U}_0^{bdd}(H),$$

so $\mathcal{U}_0^{bdd}(H)$ takes either the form $(\sup_{\mathbb{P}^* \in \mathcal{D}} \mathbb{E}^*[H], \infty)$, or $[\sup_{\mathbb{P}^* \in \mathcal{D}} \mathbb{E}^*[H], \infty)$.

The following result shows that for α tending to 1, the bounded superhedging price can be approximated by a similar notion of bounded α -quantile hedging price. For $C > 0$ and $\alpha \in (0, 1]$, we define

$$\mathcal{U}_0^{C, \alpha}(H) := \left\{ u \in \mathbb{R} : \exists \xi \in \Xi \text{ s.t. } \max_{k=1, \dots, T} \|\xi_k\|_\infty \leq C, \right. \\ \left. \text{and } \mathbb{P} \left[u + \sum_{k=1}^T \xi_k \cdot (X_k - X_{k-1}) \geq H \right] \geq \alpha \right\},$$

and let $\mathcal{U}_0^C(H) := \mathcal{U}_0^{C, 1}(H)$. We note that $\mathcal{U}_0^C(H) \subseteq \mathcal{U}_0^{bdd}(H)$ for any $C > 0$.

5.2. Approximation of the superhedging price at time $t = 0$

Lemma 5.7. *Suppose Assumption 5.5 holds. Then for any $\varepsilon > 0$, there exists $C \in (0, \infty)$ such that $\lim_{\alpha \rightarrow 1} \inf \mathcal{U}_0^{C, \alpha}(H)$ exists and*

$$\inf \mathcal{U}_0^{bdd}(H) \leq \lim_{\alpha \rightarrow 1} \inf \mathcal{U}_0^{C, \alpha}(H) \leq \inf \mathcal{U}_0^{bdd}(H) + \varepsilon.$$

Proof. Fix $\varepsilon > 0$. By Remark 5.6, there exists a predictable strategy $\tilde{\xi}$ such that $\max_{k=1, \dots, T} \|\tilde{\xi}_k\|_\infty < \infty$ and

$$\inf \mathcal{U}_0^{bdd} + \frac{\varepsilon}{2} + \sum_{k=1}^T \tilde{\xi}_k \cdot (X_k - X_{k-1}) \geq H \quad \mathbb{P}\text{-a.s.} \quad (5.3)$$

We define $C = C(\varepsilon)$ by

$$C := \max_{k=1, \dots, T} \|\tilde{\xi}_k\|_\infty + 1.$$

By (5.3) and by definition of C , we have $(\inf \mathcal{U}_0^{bdd} + \frac{\varepsilon}{2}) \in \mathcal{U}_0^C$, and therefore

$$\inf \mathcal{U}_0^C \leq \inf \mathcal{U}_0^{bdd} + \frac{\varepsilon}{2}. \quad (5.4)$$

Let $(\alpha_n)_{n \in \mathbb{N}} \subset (0, 1)$ be an increasing sequence with $\lim_{n \rightarrow \infty} \alpha_n = 1$. Then $\mathcal{U}_0^{C, \alpha_{n+1}} \subseteq \mathcal{U}_0^{C, \alpha_n}$, implying

$$\inf \mathcal{U}_0^{C, \alpha_n} \leq \inf \mathcal{U}_0^{C, \alpha_{n+1}} \leq \inf \mathcal{U}_0^{C, 1} = \inf \mathcal{U}_0^C. \quad (5.5)$$

Hence the sequence $(u_n)_{n \in \mathbb{N}}$ given by $u_n := \inf \mathcal{U}_0^{C, \alpha_n}$ is increasing and bounded, so the limit

$$u^C := \lim_{n \rightarrow \infty} u_n = \lim_{n \rightarrow \infty} \inf \mathcal{U}_0^{C, \alpha_n}$$

exists and satisfies $u^C \leq \inf \mathcal{U}_0^C$ by (5.5). Thus, (5.4) yields

$$\lim_{\alpha \rightarrow 1} \inf \mathcal{U}_0^{C, \alpha} = \lim_{n \rightarrow \infty} \inf \mathcal{U}_0^{C, \alpha_n} \leq \inf \mathcal{U}_0^C \leq \inf \mathcal{U}_0^{bdd} + \varepsilon.$$

It remains to show that $\inf \mathcal{U}_0^{bdd} \leq \lim_{\alpha \rightarrow 1} \inf \mathcal{U}_0^{C, \alpha}$. Let $\delta > 0$. Since the sequence $(u_n)_{n \in \mathbb{N}}$ is bounded, in particular $\mathcal{U}_0^{C, \alpha_n}$ is nonempty for each $n \in \mathbb{N}$. Moreover, because $\mathcal{U}_0^{C, \alpha_n}$ is an interval, for every $n \in \mathbb{N}$ there exists a predictable strategy $\xi^{(n)}$ such that $\max_{k=1, \dots, T} \|\xi_k^{(n)}\|_\infty \leq C$ and

$$\mathbb{P} \left[u_n + \delta + \sum_{k=1}^T \xi_k^{(n)} \cdot (X_k - X_{k-1}) \geq H \right] \geq \alpha_n, \quad (5.6)$$

i.e., $(u_n + \delta) \in \mathcal{U}_0^{C, \alpha_n}$. For $n \in \mathbb{N}$, define $A_n \in \mathcal{F}_T$ by

$$A_n := \left\{ \omega \in \Omega : u_n + \delta + \sum_{k=1}^T \xi_k^{(n)}(\omega) \cdot (X_k(\omega) - X_{k-1}(\omega)) \geq H(\omega) \right\}.$$

5.2. Approximation of the superhedging price at time $t = 0$

Then $\mathbb{P}[A_n] \geq \alpha_n$ by (5.6), and hence $\lim_{n \rightarrow \infty} \mathbb{P}[A_n] = 1$. Since $\xi^{(n)}$ is bounded, by part (i) of Theorem 2.21 the value process

$$V_t^{(n)} := u_n + \delta + \sum_{k=1}^t \xi_k^{(n)} \cdot (X_k - X_{k-1})$$

is a $\mathcal{P}$ -martingale for all $n \in \mathbb{N}$. Let $\mathbb{P}^* \in \mathcal{P}$. Using the martingale property we have $V_0^{(n)} = \mathbb{E}^*[V_T^{(n)}]$, and therefore

$$\begin{aligned} u_n + \delta &= \mathbb{E}^* \left[u_n + \delta + \sum_{k=1}^T \xi_k^{(n)} \cdot (X_k - X_{k-1}) \right] \\ &\geq \mathbb{E}^*[H \mathbb{1}_{A_n}] + \mathbb{E}^* \left[\left(u_n + \delta + \sum_{k=1}^T \xi_k^{(n)} \cdot (X_k - X_{k-1}) \right) \mathbb{1}_{A_n^C} \right] \\ &\geq \mathbb{E}^*[H \mathbb{1}_{A_n}] + \mathbb{E}^* \left[\left(u_n + \delta - \sum_{k=1}^T \sum_{i=1}^d |\xi_k^{i,(n)}| |X_k^i - X_{k-1}^i| \right) \mathbb{1}_{A_n^C} \right] \\ &\geq \mathbb{E}^*[H \mathbb{1}_{A_n}] + \mathbb{E}^* \left[\left(u_n + \delta - C \sum_{k=1}^T \sum_{i=1}^d |X_k^i - X_{k-1}^i| \right) \mathbb{1}_{A_n^C} \right]. \quad (5.7) \end{aligned}$$

Recalling that $u_n \leq u^C \leq \inf \mathcal{U}_0^{bdd} + \varepsilon = \inf \mathcal{U}_0 + \varepsilon = \text{const.}$ by (5.2), we get

$$\left| u_n + \delta - C \sum_{k=1}^T \sum_{i=1}^d |X_k^i - X_{k-1}^i| \right| \leq K + |C| \sum_{k=1}^T \sum_{i=1}^d (|X_k^i| + |X_{k-1}^i|),$$

for some constant $K > 0$. Since X is a d -dimensional $\mathbb{P}^*$ -martingale, this implies

$$\mathbb{E}^* \left[\left| u_n + \delta - C \sum_{k=1}^T \sum_{i=1}^d |X_k^i - X_{k-1}^i| \right| \right] < \infty.$$

Moreover, for any $\eta \in (0, 1)$, by (5.6) we have

$$\mathbb{P}[|\mathbb{1}_{A_n} - 1| > \eta] = \mathbb{P}[\mathbb{1}_{A_n^C} > \eta] = \mathbb{P}[\mathbb{1}_{A_n^C}] \xrightarrow{n \rightarrow \infty} 0,$$

so $\mathbb{1}_{A_n} \xrightarrow{n \rightarrow \infty} 1$ in probability. Thus by dominated convergence we obtain

$$\lim_{n \rightarrow \infty} \mathbb{E}^*[H \mathbb{1}_{A_n}] = \mathbb{E}^*[H] \quad (5.8)$$

and

$$\lim_{n \rightarrow \infty} \mathbb{E}^* \left[\left(u_n + \delta - C \sum_{k=1}^T \sum_{i=1}^d |X_k^i - X_{k-1}^i| \right) \mathbb{1}_{A_n^C} \right] = 0. \quad (5.9)$$

5.2. Approximation of the superhedging price at time $t = 0$

Letting $n \rightarrow \infty$, (5.7) together with (5.8) and (5.9) yields

$$\begin{aligned} u^C + \delta &\geq \lim_{n \rightarrow \infty} \left(\mathbb{E}^*[H \mathbb{1}_{A_n}] + \mathbb{E}^* \left[\left(u_n + \delta - C \sum_{k=1}^T \sum_{i=1}^d |X_k^i - X_{k-1}^i| \right) \mathbb{1}_{A_n^C} \right] \right) \\ &= \mathbb{E}^*[H]. \end{aligned}$$

Since $\mathbb{P}^* \in \mathcal{P}$ was arbitrary, it follows that

$$\sup_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[H] \leq u^C + \delta = \lim_{n \rightarrow \infty} \inf \mathcal{U}_0^{C, \alpha_n} + \delta.$$

Because $\delta < 0$ was arbitrary and by Assumption 5.5, we get

$$\inf \mathcal{U}_0^{bdd} = \inf \mathcal{U}_0 = \sup_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[H] \leq \lim_{\alpha \rightarrow 1} \inf \mathcal{U}_0^{C, \alpha},$$

which concludes the proof. $\square$

Finally, we prove the main approximation result of this work, following Biagini *et al.* [1, Theorem 3.14]. Under Assumption 5.5, we show that the superhedging price of H can be approximated arbitrarily well by the truncated approximate α -quantile hedging price $\mathcal{U}^{\Theta, C, \alpha}(H)$, for adequate $C > 0$ and $\alpha \in (0, 1)$. We note that Assumption 5.5 is necessary for Theorem 5.8 to hold, as shown in Biagini *et al.* [1, Example 3.15]. However, it represents a relatively mild assumption, as it holds for a large class of models (see also [1, Remark 3.12]).

Theorem 5.8. *Assume σ is bounded and nonconstant and Assumption 5.5 holds. Then for any $\varepsilon > 0$ there exists $\alpha_0 \in (0, 1)$ and $C \in (0, \infty)$ such that*

$$\inf \mathcal{U}_0(H) - \varepsilon \leq \inf \mathcal{U}_0^{\Theta, C, \alpha}(H) \leq \inf \mathcal{U}_0(H) + \varepsilon \quad \text{for all } \alpha \in (\alpha_0, 1).$$

Proof. Let $\varepsilon > 0$. As in the proof of Lemma 5.7, there exists a predictable strategy $\tilde{\xi} \in \Theta$ such that $\max_{k=1, \dots, T} \|\tilde{\xi}_k\|_\infty < \infty$ and

$$\inf \mathcal{U}_0^{bdd} + \frac{\varepsilon}{2} + \sum_{k=1}^T \tilde{\xi}_k \cdot (X_k - X_{k-1}) \geq H \quad \mathbb{P}\text{-a.s.} \quad (5.10)$$

Again, we define $C = C(\varepsilon)$ by

$$C := \max_{k=1, \dots, T} \|\tilde{\xi}_k\|_\infty + 1.$$

We note that $\mathcal{U}_0^{\Theta, C, \alpha} \subseteq \mathcal{U}_0^{C, \alpha}$ for all $\alpha \in (0, 1)$: indeed, for $u \in \mathcal{U}_0^{\Theta, C, \alpha}$ with corresponding network parameters $\theta_{k-1, \xi} \in \Theta_{k-1}$, $k = 1, \dots, T$, we can define $\xi \in \Xi$ by $\xi_k := F^{\theta_{k-1, \xi}, C}(\mathbf{Y}_{k-1})$, $k = 1, \dots, T$. Then we have $\max_{k=1, \dots, T} \|\xi_k\|_\infty \leq C$ by definition of $F^{\theta_{k-1, \xi}, C}$ (see equation (5.2)) and

$$\mathbb{P} \left[u + \sum_{k=1}^T \xi_k \cdot (X_k - X_{k-1}) \geq H \right] \geq \alpha,$$

5.2. Approximation of the superhedging price at time $t = 0$

so $u \in \mathcal{U}_0^{C,\alpha}$. In particular, we get $\inf \mathcal{U}_0^{C,\alpha} \leq \inf \mathcal{U}_0^{\Theta,C,\alpha}$. Furthermore, by Lemma 5.7 there exists $\alpha_0 = \alpha_0(\varepsilon)$ such that for all $\alpha \in (\alpha_0, 1)$

$$\inf \mathcal{U}_0^{bdd} - \varepsilon \leq \inf \mathcal{U}_0^{C,\alpha}.$$

Hence, using Assumption 5.5, for all $\alpha \in (\alpha_0, 1)$ we obtain

$$\inf \mathcal{U}_0 - \varepsilon = \inf \mathcal{U}_0^{bdd} - \varepsilon \leq \inf \mathcal{U}_0^{C,\alpha} \leq \inf \mathcal{U}_0^{\Theta,C,\alpha}. \quad (5.11)$$

It remains to show that for all $\alpha \in (\alpha_0, 1)$,

$$\inf \mathcal{U}_0^{\Theta,C,\alpha} \leq \inf \mathcal{U}_0 + \varepsilon.$$

Fix $\alpha \in (\alpha_0, 1)$. For $n \in \mathbb{N}$, we define

$$A_n := \{\omega \in \Omega : \|X_k - X_{k-1}\| \leq n \text{ for } k = 1, \dots, T\}$$

and

$$M_n := \left\{ \omega \in \Omega : \inf \mathcal{U}_0^{bdd} + \frac{\varepsilon}{2} + \sum_{k=1}^T \tilde{\xi}_k(\omega) \cdot (X_k(\omega) - X_{k-1}(\omega)) \geq H \right\} \cap A_n.$$

Then $M_n \subseteq M_{n+1}$ for each $n \in \mathbb{N}$, so using (5.10) and by continuity from below we get

$$1 = \mathbb{P} \left[\inf \mathcal{U}_0^{bdd} + \frac{\varepsilon}{2} + \sum_{k=1}^T \tilde{\xi}_k \cdot \Delta X_k \geq H \right] = \mathbb{P} \left[\bigcup_{n \in \mathbb{N}} M_n \right] = \lim_{n \rightarrow \infty} \mathbb{P}[M_n].$$

Hence there exists $n_0 \in \mathbb{N}$ such that

$$\mathbb{P}[M_{n_0}] \geq \frac{\alpha + 1}{2}. \quad (5.12)$$

We recall that $\tilde{\xi}$ is predictable and $\mathcal{F}_t = \sigma(\mathbf{Y}_{\mathbf{t}})$ for $t = 0, \dots, T$. Hence, for each $t = 1, \dots, T$ there exists a measurable function $f_{t-1} : \mathbb{R}^{mt} \rightarrow \mathbb{R}^d$ such that $\tilde{\xi}_t = f_{t-1}(\mathbf{Y}_{\mathbf{t}-1})$. For $t = 1, \dots, T$, let μ_t be the probability measure given by the distribution of $\mathbf{Y}_{\mathbf{t}-1}$ under $\mathbb{P}$, i.e., $\mu_t(B) = \mathbb{P}[\{\omega \in \Omega : \mathbf{Y}_{\mathbf{t}-1}(\omega) \in B\}]$ for $B \in \mathcal{B}(\mathbb{R}^{mt})$. Then, by Corollary 5.3, for each $t = 1, \dots, T$ there exists $\theta_{t-1, \tilde{\xi}} \in \Theta_{t-1}$ such that

$$\begin{aligned} \frac{1 - \alpha}{2T} &> \mu_t \left[\left\{ x \in \mathbb{R}^{mt} : \|f_{t-1}(x) - F^{\theta_{t-1, \tilde{\xi}}}(x)\| > \left(\frac{\varepsilon}{2n_0 T} \wedge \frac{1}{2} \right) \right\} \right] \\ &= \mathbb{P}[D_t], \end{aligned} \quad (5.13)$$

where

$$D_t := \left\{ \omega \in \Omega : \|f_{t-1}(\mathbf{Y}_{\mathbf{t}-1}(\omega)) - F^{\theta_{t-1, \tilde{\xi}}}(\mathbf{Y}_{\mathbf{t}-1}(\omega))\| > \left(\frac{\varepsilon}{2n_0 T} \wedge \frac{1}{2} \right) \right\}.$$

5.2. Approximation of the superhedging price at time $t = 0$

For $t = 1, \dots, T$, define

$$\tilde{D}_t := \left\{ \omega \in \Omega : \|f_{t-1}(\mathbf{Y}_{t-1}(\omega)) - F^{\theta_{t-1}, \tilde{\xi}, C}(\mathbf{Y}_{t-1}(\omega))\| > \left(\frac{\varepsilon}{2n_0 T} \wedge \frac{1}{2} \right) \right\}.$$

We show that $D_t = \tilde{D}_t$ for every t . For $\omega \in D_t^C$ and $i = 1, \dots, d$, we have

$$\begin{aligned} \left| F_i^{\theta_{t-1}, \tilde{\xi}}(\mathbf{Y}_{t-1}(\omega)) \right| &\leq \left\| F_i^{\theta_{t-1}, \tilde{\xi}}(\mathbf{Y}_{t-1}(\omega)) \right\| \leq \|\tilde{\xi}_k\|_\infty + \left(\frac{\varepsilon}{2n_0 T} \wedge \frac{1}{2} \right) \\ &< \|\tilde{\xi}_k\|_\infty + 1 \leq C \end{aligned}$$

by the definition of C . Therefore $F^{\theta_{t-1}, \tilde{\xi}, C}(\mathbf{Y}_{t-1}) = F^{\theta_{t-1}, \tilde{\xi}}(\mathbf{Y}_{t-1})$ on D_t^C , implying $D_t^C \subseteq \tilde{D}_t^C$. Similarly, for $\omega \in \tilde{D}_t^C$ and $i = 1, \dots, d$ we get

$$\left| F_i^{\theta_{t-1}, \tilde{\xi}, C}(\mathbf{Y}_{t-1}(\omega)) \right| \leq \left\| F_i^{\theta_{t-1}, \tilde{\xi}, C}(\mathbf{Y}_{t-1}(\omega)) \right\| \leq \|\tilde{\xi}_k\|_\infty + \left(\frac{\varepsilon}{2n_0 T} \wedge \frac{1}{2} \right) < C,$$

so it follows that $\tilde{D}_t^C \subseteq D_t^C$, and thus $D_t = \tilde{D}_t$. In particular, by (5.13) this implies

$$\mathbb{P}[\tilde{D}_t^C] = 1 - \mathbb{P}[D_t] \geq 1 - \frac{1 - \alpha}{2T}. \quad (5.14)$$

For simplicity of notation, let $F_i^{\theta_{t-1}, \tilde{\xi}, C} := F_i^{\theta_{t-1}, \tilde{\xi}, C}(\mathbf{Y}_{t-1})$, $f_{k-1} := f_{k-1}(\mathbf{Y}_{k-1})$ and $\Delta X_k := (X_k - X_{k-1})$. By definition, on $A_{n_0} \cap \tilde{D}_1^C \cap \dots \cap \tilde{D}_T^C$ we have

$$\begin{aligned} \sum_{k=1}^T \tilde{\xi}_k \cdot \Delta X_k &= \sum_{k=1}^T f_{k-1} \cdot \Delta X_k \\ &= \sum_{k=1}^T (f_{k-1} - F^{\theta_{k-1}, \tilde{\xi}, C}) \cdot \Delta X_k + \sum_{k=1}^T F^{\theta_{k-1}, \tilde{\xi}, C} \cdot \Delta X_k \\ &\leq \sum_{k=1}^T \|f_{k-1} - F^{\theta_{k-1}, \tilde{\xi}, C}\| \|\Delta X_k\| + \sum_{k=1}^T F^{\theta_{k-1}, \tilde{\xi}, C} \cdot \Delta X_k \\ &\leq n_0 \sum_{k=1}^T \|f_{k-1} - F^{\theta_{k-1}, \tilde{\xi}, C}\| + \sum_{k=1}^T F^{\theta_{k-1}, \tilde{\xi}, C} \cdot \Delta X_k \\ &\leq n_0 \left(T \frac{\varepsilon}{2n_0 T} \right) + \sum_{k=1}^T F^{\theta_{k-1}, \tilde{\xi}, C} \cdot \Delta X_k \\ &= \frac{\varepsilon}{2} + \sum_{k=1}^T F^{\theta_{k-1}, \tilde{\xi}, C} \cdot \Delta X_k, \end{aligned}$$

5.3. Approximation of the superhedging price at time $t > 0$

where we used the Cauchy-Schwarz inequality in the third step. Then, by definition of M_n , on $M_{n_0} \cap \tilde{D}_1^C \cap \dots \cap \tilde{D}_T^C$ we have

$$\begin{aligned} H &\leq \inf \mathcal{U}_0^{bdd} + \frac{\varepsilon}{2} + \sum_{k=1}^T \tilde{\xi}_k \cdot \Delta X_k \\ &\leq \inf \mathcal{U}_0^{bdd} + \frac{\varepsilon}{2} + \frac{\varepsilon}{2} + \sum_{k=1}^T F^{\theta_{k-1}, \tilde{\xi}, C} \cdot \Delta X_k \\ &= \inf \mathcal{U}_0^{bdd} + \varepsilon + \sum_{k=1}^T F^{\theta_{k-1}, \tilde{\xi}, C} \cdot \Delta X_k. \end{aligned}$$

Therefore, using the Fréchet inequalities, (5.12) and (5.14), we obtain

$$\begin{aligned} \mathbb{P} \left[\inf \mathcal{U}_0^{bdd} + \varepsilon + \sum_{k=1}^T F^{\theta_{k-1}, \tilde{\xi}, C} \cdot \Delta X_k \geq H \right] &\geq \mathbb{P}[M_{n_0} \cap \tilde{D}_1^C \cap \dots \cap \tilde{D}_T^C] \\ &\geq \mathbb{P}[M_{n_0}] + \sum_{k=1}^T \mathbb{P}[\tilde{D}_k^C] - T \\ &\geq \frac{\alpha + 1}{2} + T \left(1 - \frac{1 - \alpha}{2T} \right) - T \\ &= \alpha. \end{aligned}$$

Thus, we conclude that $(\inf \mathcal{U}_0^{bdd} + \varepsilon) \in \mathcal{U}_0^{\Theta, C, \alpha}$, which by Assumption 5.5 implies

$$\inf \mathcal{U}_0^{\Theta, C, \alpha} \leq \inf \mathcal{U}_0^{bdd} + \varepsilon = \inf \mathcal{U}_0 + \varepsilon.$$

Since $\alpha \in (\alpha_0, 1)$ was arbitrary, together with (5.11) this concludes the proof. $\square$

5.3 Approximation of the superhedging price at time $t > 0$

In Section 3.3 we saw that the upper Snell envelope $U^\uparrow$ of a European contingent claim H represents the minimal investment to start a superhedging strategy at all times $t > 0$. Moreover, we provided a characterization for the consumption process B in the optional decomposition of $U^\uparrow$. In this final section, we approximate B by neural networks.

Let H denote a discounted European contingent claim. We recall the set of initial investments for the purchase of a superhedging strategy for H at time $t = 1, \dots, T$ is given by

$$\mathcal{U}_t(H) := \left\{ U_t \in L^0(\Omega, \mathcal{F}_t, \mathbb{P}) : \exists \xi \in \Xi \text{ s.t. } U_t + \sum_{k=t+1}^T \xi_k \cdot \Delta X_k \geq H \text{ } \mathbb{P}\text{-a.s.} \right\}.$$

5.3. Approximation of the superhedging price at time $t > 0$

By Theorem 3.23, the upper Snell envelope of H

$$U_t^\uparrow := \operatorname{ess\,sup}_{\mathbb{P}^* \in \mathcal{D}} \mathbb{E}^*[H | \mathcal{F}_t], \quad t = 1, \dots, T,$$

satisfies $U_t^\uparrow \in \mathcal{U}_t(H)$ and

$$U_t^\uparrow = \operatorname{ess\,inf} \mathcal{U}_t(H) \quad \mathbb{P}\text{-a.s.}, \text{ for } t = 1, \dots, T.$$

Moreover, by the optional decomposition theorem there exists a d -dimensional predictable process $\xi \in \Xi$ and an adapted increasing process B with $B_0 = 0$ such that

$$U_t^\uparrow = U_0^\uparrow + \sum_{k=1}^t \xi_k \cdot (X_k - X_{k-1}) - B_t \quad \mathbb{P}\text{-a.s. for } t = 1, \dots, T. \quad (5.15)$$

Theorem 5.8 in the previous section showed that we can approximate the time-0 superhedging price $\inf \mathcal{U}_0(H) = U_0^\uparrow$ by feedforward neural networks. In the following, our goal is to approximate the process of consumption B . We recall that by Theorem 3.24, for $t = 1, \dots, T$, the process B satisfies

$$B_t = \operatorname{ess\,sup} \mathcal{B}_t \quad \mathbb{P}\text{-a.s.},$$

where the set $\mathcal{B}_t$ is given by

$$\mathcal{B}_t := \left\{ D_t \in L^0(\Omega, \mathcal{F}_t, \mathbb{P}) : B_{t-1} \leq D_t \leq U_0^\uparrow + \sum_{k=1}^t \xi_k \cdot \Delta X_k - H \text{ } \mathbb{P}\text{-a.s.} \right\}.$$

For $\varepsilon, \delta \in (0, 1)$, we define the set

$$\mathcal{B}_t^{\Theta_t, \varepsilon, \delta} = \left\{ F^{\theta_t} : \theta_t \in \Theta_t \text{ and } \mathbb{P} \left[B_{t-1} - \delta \leq F^{\theta_t} \leq U_0^\uparrow + \sum_{k=1}^t \xi_k \cdot \Delta X_k - H + \delta \right] > 1 - \varepsilon \right\}.$$

The following result is the first step towards constructing an approximation of the consumption process B by neural networks.

Proposition 5.9. *Assume σ is bounded and nonconstant. Then for any $\varepsilon, \delta > 0$ there exist neural networks $F^{\theta_0, \varepsilon, \delta}, \dots, F^{\theta_T, \varepsilon, \delta}$ such that for all $t = 0, \dots, T$,*

$$F^{\theta_t, \varepsilon, \delta} \in \mathcal{B}_t^{\Theta_t, \varepsilon, \delta} \quad \text{and} \quad \mathbb{P} \left[\left| F^{\theta_t, \varepsilon, \delta} - B_t \right| > \delta \right] < \varepsilon. \quad (5.16)$$

In particular, there exists a sequence of neural networks $(F^{\theta_0^n}, \dots, F^{\theta_T^n})_{n \in \mathbb{N}}$ with $F^{\theta_t^n} \in \mathcal{B}_t^{\Theta_t, \frac{1}{n}, \frac{1}{n}}$ for all $n \in \mathbb{N}$ and $t = 0, \dots, T$, such that

$$(F^{\theta_0^n}, \dots, F^{\theta_T^n}) \xrightarrow{n \rightarrow \infty} (B_0, \dots, B_T) \quad \mathbb{P}\text{-a.s.}$$

5.3. Approximation of the superhedging price at time $t > 0$

Proof. Recalling that $B_0 = 0$, for $t = 0$ the statement is clear. Fix $\varepsilon, \delta > 0$ and $t \in \{1, \dots, T\}$. We note that the set $\mathcal{B}_t$ is directed upwards; indeed, for any $B_t^1, B_t^2 \in \mathcal{B}_t$ we have by definition that $B_t^1 \vee B_t^2 \in \mathcal{B}_t$. Hence by Theorem 2.10 there exists an increasing sequence $(B_t^k)_{k \in \mathbb{N}} \subseteq \mathcal{B}_t$ such that

$$B_t^k \xrightarrow{k \rightarrow \infty} \text{ess sup } \mathcal{B}_t = B_t \quad \mathbb{P}\text{-a.s.}$$

Since almost sure convergence implies convergence in probability, there exists $K = K(\varepsilon, \delta) \in \mathbb{N}$ such that

$$\mathbb{P}\left[|B_t^k - B_t| > \frac{\delta}{2}\right] < \frac{\varepsilon}{2}, \quad \text{for all } k \geq K. \quad (5.17)$$

Let $k \geq K$. As $\mathcal{F}_t = \sigma(\mathbf{Y}_t)$, there exists a measurable function

$$f_t^k : (\mathbb{R}^{m(t+1)}, \mathcal{B}(\mathbb{R}^{m(t+1)})) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$$

such that $B_t^k = f_t^k(\mathbf{Y}_t)$. Let μ_t be the probability measure given by the distribution of $\mathbf{Y}_t$ under $\mathbb{P}$. Then, by Corollary 5.3, there exists $\theta_t^k \in \Theta_t$ such that

$$\mathbb{P}\left[|f_t^k(\mathbf{Y}_t) - F^{\theta_t^k}(\mathbf{Y}_t)| > \frac{\delta}{2}\right] = \mu_t\left[x \in \mathbb{R}^{m(t+1)} : |f_t^k(x) - F^{\theta_t^k}(x)| > \frac{\delta}{2}\right] < \frac{\varepsilon}{2}.$$

From the triangle inequality and de Morgan's law, we get

$$\begin{aligned} & \left\{ \omega \in \Omega : |B_t(\omega) - F^{\theta_t^k}(\mathbf{Y}_t(\omega))| > \delta \right\} \\ & \subseteq \left\{ \omega \in \Omega : |B_t(\omega) - B_t^k(\omega)| + |B_t^k(\omega) - F^{\theta_t^k}(\mathbf{Y}_t(\omega))| > \delta \right\} \\ & \subseteq \left\{ \omega \in \Omega : |B_t(\omega) - B_t^k(\omega)| > \frac{\delta}{2} \right\} \cup \left\{ \omega \in \Omega : |B_t^k(\omega) - F^{\theta_t^k}(\mathbf{Y}_t(\omega))| > \frac{\delta}{2} \right\}, \end{aligned}$$

which by subadditivity implies

$$\begin{aligned} \mathbb{P}[|B_t - F^{\theta_t^k}(\mathbf{Y}_t)| > \delta] & \leq \mathbb{P}\left[|B_t - B_t^k| > \frac{\delta}{2}\right] + \mathbb{P}\left[|B_t^k - F^{\theta_t^k}(\mathbf{Y}_t)| > \frac{\delta}{2}\right] \\ & < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \\ & = \varepsilon. \end{aligned}$$

We show that $F^{\theta_t^k}(\mathbf{Y}_t) \in \mathcal{B}_t^{\Theta, \varepsilon, \delta}$. Using that B is increasing, $U_t^\uparrow \in \mathcal{U}_t$ and (5.15) we have

$$B_{t-1} \leq B_t \leq U_0^\uparrow + \sum_{k=1}^T \xi_k \cdot \Delta X_k - H \quad \mathbb{P}\text{-a.s.}$$

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Therefore we obtain

$$\begin{aligned} 1 - \varepsilon &< \mathbb{P}\left[\left|B_t - F_t^{\theta^k}(\mathbf{Y}_t)\right| \leq \delta\right] \\ &= \mathbb{P}\left[B_t - \delta \leq F_t^{\theta^k}(\mathbf{Y}_t) \leq B_t + \delta\right] \\ &\leq \mathbb{P}\left[B_{t-1} - \delta \leq F_t^{\theta^k}(\mathbf{Y}_t) \leq U_0^\uparrow + \sum_{k=1}^T \xi_k \cdot \Delta X_k - H\right], \end{aligned}$$

thus $F_t^{\theta^k}(\mathbf{Y}_t) \in \mathcal{B}_t^{\Theta, \varepsilon, \delta}$. Since $k \geq K$ and $t \in \{1, \dots, T\}$ were arbitrary, we can choose any $k^* \geq K$ and letting $F_t^{\theta_t, \varepsilon, \delta} := F_t^{\theta_t^*}(\mathbf{Y}_t)$, for $t = 1, \dots, T$, proves (5.16). For the final part of the proof, let $\varepsilon = \delta = \frac{1}{m}$ for $m \in \mathbb{N}$. By the first part of Proposition 5.9, there exist neural networks $F^{\theta_0, \frac{1}{m}, \frac{1}{m}}, \dots, F^{\theta_T, \frac{1}{m}, \frac{1}{m}}$ such that for all $t = 0, \dots, T$,

$$F^{\theta_t, \frac{1}{m}, \frac{1}{m}} \in \mathcal{B}^{\Theta_t, \frac{1}{m}, \frac{1}{m}} \quad \text{and} \quad \mathbb{P}\left[\left|F^{\theta_t, \frac{1}{m}, \frac{1}{m}} - B_t\right| > \frac{1}{m}\right] < \frac{1}{m}. \quad (5.18)$$

In particular, this implies that $F^{\theta_t, \frac{1}{m}, \frac{1}{m}} \xrightarrow{m \rightarrow \infty} B_t$ in probability, for all $t = 0, \dots, T$. By passing to a suitable subsequence $(m_n)_{n \in \mathbb{N}}$, we get

$$F^{\theta_t, \frac{1}{m_n}, \frac{1}{m_n}} \xrightarrow{n \rightarrow \infty} B_t \quad \mathbb{P}\text{-a.s.},$$

simultaneously for all $t = 0, \dots, T$. Thus letting

$$F_t^{\theta^n} := F^{\theta_t, \frac{1}{m_n}, \frac{1}{m_n}}, \quad t = 0, \dots, T, n \in \mathbb{N},$$

concludes the proof. $\square$

Finally, we prove the main approximation result of this section. Let $\delta > 0$. Recursively, we define the set

$$\begin{aligned} \tilde{\mathcal{B}}_t^{\Theta, \delta} &:= \left\{ F^{\theta_t} \mathbb{1}_A + B_{t-1}^{\Theta, \delta} \mathbb{1}_{A^c} : \theta_t \in \Theta_t, A \in \mathcal{F}_t \text{ and} \right. \\ &\quad \left. B_{t-1}^{\Theta, \delta} \leq F^{\theta_t} \mathbb{1}_A + B_{t-1}^{\Theta, \delta} \mathbb{1}_{A^c} \leq U_0^\uparrow + \sum_{k=1}^T \xi_k \cdot \Delta X_k - H + \delta \text{ } \mathbb{P}\text{-a.s.} \right\} \end{aligned}$$

for $t = 1, \dots, T$, and the *approximated process of consumption* by

$$B_0^{\Theta, \delta} := 0 \quad \text{and} \quad B_t^{\Theta, \delta} := \text{ess sup } \tilde{\mathcal{B}}_t^{\Theta, \delta}, \quad \text{for } t = 1, \dots, T.$$

Theorem 5.10. *Assume σ is bounded and nonconstant and let $\delta > 0$. Then*

$$\left| B_t^{\Theta, \delta} - B_t \right| \leq \delta \quad \mathbb{P}\text{-a.s.}, \text{ for } t = 0, \dots, T.$$

5.3. Approximation of the superhedging price at time $t > 0$

Proof. We prove the statement by induction. For $t = 0$, we have $B_0 = B_0^{\Theta, \delta}$ by definition. Assume

$$\left| B_{t-1}^{\Theta, \delta} - B_{t-1} \right| \leq \delta \quad \mathbb{P}\text{-a.s.}$$

for some $t \in \{1, \dots, T\}$. Note that for all $s = 1, \dots, T$, $B_s^{\Theta, \delta} \geq B_{s-1}^{\Theta, \delta}$ and since $B_0^{\Theta, \delta} = 0$, in particular $B_s^{\Theta, \delta} \geq 0$ for $s = 1, \dots, T$. Let $\theta_t \in \Theta_t$ and $A \in \mathcal{F}_t$ be such that

$$0 \leq B_{t-1}^{\Theta, \delta} \leq F^{\theta_t} \mathbb{1}_A + B_{t-1}^{\Theta, \delta} \mathbb{1}_{A^c} \leq U_0^\uparrow + \sum_{k=1}^T \xi_k \cdot \Delta X_k - H + \delta \quad \mathbb{P}\text{-a.s.} \quad (5.19)$$

We recall that by Lemma 3.25,

$$B_t + \delta = \text{ess sup } \mathcal{B}_t^\delta \quad \mathbb{P}\text{-a.s.}, \quad (5.20)$$

where

$$\mathcal{B}_t^\delta := \left\{ D_t \in L^0(\Omega, \mathcal{F}_t, \mathbb{P}) : -\delta \leq D_t - \delta \leq U_0^\uparrow + \sum_{k=1}^T \xi_k \cdot \Delta X_k - H \text{ } \mathbb{P}\text{-a.s.} \right\}.$$

By (5.19), we have

$$-\delta \leq F^{\theta_t} \mathbb{1}_A + B_{t-1}^{\Theta, \delta} \mathbb{1}_{A^c} - \delta \leq U_0^\uparrow + \sum_{k=1}^T \xi_k \cdot \Delta X_k - H \quad \mathbb{P}\text{-a.s.},$$

implying $F^{\theta_t} \mathbb{1}_A + B_{t-1}^{\Theta, \delta} \mathbb{1}_{A^c} \in \mathcal{B}_t^\delta$ and therefore

$$F^{\theta_t} \mathbb{1}_A + B_{t-1}^{\Theta, \delta} \mathbb{1}_{A^c} \leq \text{ess sup } \mathcal{B}_t^\delta = B_t + \delta \quad \mathbb{P}\text{-a.s.}$$

Since $\theta_t \in \Theta_t$ and $A \in \mathcal{F}_t$ were arbitrary, we conclude

$$B_t^{\Theta, \delta} \leq B_t + \delta \quad \mathbb{P}\text{-a.s.} \quad (5.21)$$

It remains to show

$$B_t - \delta \leq B_t^{\Theta, \delta} \quad \mathbb{P}\text{-a.s.} \quad (5.22)$$

Fix $\varepsilon \in (0, 1)$. By Proposition 5.9, there exists a neural network $F^{\theta_t, \varepsilon, \delta} \in \mathcal{B}_t^{\Theta, \varepsilon, \delta}$ such that

$$\mathbb{P}[|F^{\theta_t, \varepsilon, \delta} - B_t| > \delta] < \varepsilon. \quad (5.23)$$

Define the sets $A_1, A_2 \in \mathcal{F}_t$ by

$$A_1 := \left\{ \omega \in \Omega : B_t(\omega) - \delta \leq F^{\theta_t, \varepsilon, \delta} \leq B_t(\omega) + \delta \right\},$$

and

$$A_2 := \left\{ \omega \in \Omega : B_{t-1}^{\Theta, \delta}(\omega) \leq F^{\theta_t, \varepsilon, \delta} \right\},$$

5.3. Approximation of the superhedging price at time $t > 0$

and let $A := A_1 \cap A_2$. Then by construction of A and using that $B_{t-1}^{\Theta, \delta} \leq B_{t-1} + \delta$ $\mathbb{P}$ -a.s. by the induction hypothesis, we have

$$\begin{aligned} B_{t-1}^{\Theta, \delta} &\leq F^{\theta_t, \varepsilon, \delta} \mathbb{1}_A + B_{t-1}^{\Theta, \delta} \mathbb{1}_{A^C} \\ &\leq (B_t + \delta) \mathbb{1}_A + (B_{t-1} + \delta) \mathbb{1}_{A^C} \\ &\leq B_t + \delta \end{aligned} \quad \mathbb{P}\text{-a.s.}$$

Moreover, since $B_t = \text{ess sup } \mathcal{B}_t$ by Theorem 3.24, we get

$$B_t + \delta \leq U_0^\uparrow + \sum_{k=1}^T \xi_k \cdot \Delta X_k - H + \delta \quad \mathbb{P}\text{-a.s.},$$

and therefore we conclude

$$(F^{\theta_t, \varepsilon, \delta} \mathbb{1}_A + B_{t-1}^{\Theta, \delta} \mathbb{1}_{A^C}) \in \tilde{\mathcal{B}}_t^{\Theta, \delta}. \quad (5.24)$$

For $\omega \in A_1 \cap A_2^C$, we have

$$F^{\theta_t, \varepsilon, \delta}(\mathbf{Y}_t(\omega)) \mathbb{1}_{A_1 \cap A_2}(\omega) + B_{t-1}^{\Theta, \delta}(\omega) \mathbb{1}_{A_1^C \cup A_2^C}(\omega) = B_{t-1}^{\Theta, \delta}(\omega) \quad (5.25)$$

and

$$B_t(\omega) - \delta \leq F^{\theta_t, \varepsilon, \delta}(\mathbf{Y}_t(\omega)) < B_{t-1}^{\Theta, \delta}(\omega) \leq B_t(\omega) + \delta, \quad (5.26)$$

by the induction hypothesis. On the other hand, for $\omega \in A_1 \cap A_2$, we have

$$F^{\theta_t, \varepsilon, \delta}(\mathbf{Y}_t(\omega)) \mathbb{1}_{A_1 \cap A_2}(\omega) + B_{t-1}^{\Theta, \delta}(\omega) \mathbb{1}_{A_1^C \cup A_2^C}(\omega) = F^{\theta_t, \varepsilon, \delta}(\mathbf{Y}_t(\omega)) \quad (5.27)$$

and

$$|F^{\theta_t, \varepsilon, \delta}(\mathbf{Y}_t(\omega)) - B_t(\omega)| \leq \delta, \quad (5.28)$$

by definition of A_1 . Since A_1 is the disjoint union of $A_1 \cap A_2$ and $A_1 \cap A_2^C$, from equations (5.25)–(5.28) we conclude that for $\omega \in A_1$,

$$\left| \left(F^{\theta_t, \varepsilon, \delta}(\mathbf{Y}_t(\omega)) \mathbb{1}_A(\omega) + B_{t-1}^{\Theta, \delta}(\omega) \mathbb{1}_{A^C}(\omega) \right) - B_t(\omega) \right| \leq \delta. \quad (5.29)$$

Since by (5.23) we have $\mathbb{P}[A_1] > 1 - \varepsilon$, (5.29) implies

$$\mathbb{P} \left[\left| \left(F^{\theta_t, \varepsilon, \delta} \mathbb{1}_A + B_{t-1}^{\Theta, \delta} \mathbb{1}_{A^C} \right) - B_t \right| > \delta \right] \leq \mathbb{P}[A_1^C] < \varepsilon.$$

Moreover, $F^{\theta_t, \varepsilon, \delta} \mathbb{1}_A + B_{t-1}^{\Theta, \delta} \mathbb{1}_{A^C} \leq B_t^{\Theta, \delta} = \text{ess sup } \tilde{\mathcal{B}}_t^{\Theta, \delta}$ by (5.24), so

$$\mathbb{P} \left[B_t^{\Theta, \delta} < B_t - \delta \right] \leq \mathbb{P} \left[F^{\theta_t, \varepsilon, \delta} \mathbb{1}_A + B_{t-1}^{\Theta, \delta} \mathbb{1}_{A^C} < B_t - \delta \right] < \varepsilon.$$

As $\varepsilon \in (0, 1)$ was arbitrary, this proves (5.22). Hence, by (5.21) we obtain

$$|B_t^{\Theta, \delta} - B_t| \leq \delta \quad \mathbb{P}\text{-a.s.},$$

which shows the induction step and thus concludes the proof. $\square$

Chapter 6

Conclusion

In this thesis, we approximated the superhedging price of a discounted European contingent claim H by neural networks. Our work was developed in an arbitrage-free incomplete financial market model in finite discrete time, with price fluctuation of the assets and trading strategies modeled by random variables on a filtered probability space $(\Omega, \mathcal{F}, \mathbb{F}, \mathbb{P})$.

First, we introduced the upper Snell envelope of the contingent claim H , and showed that it is a $\mathbb{P}^*$ -supermartingale for all equivalent martingale measures $\mathbb{P}^* \in \mathcal{P}(\mathbb{P})$. Using the optional decomposition theorem, we could then represent the upper Snell envelope as the difference of a $\mathcal{P}$ -martingale and an adapted increasing process, called process of consumption. This served as the key element for showing that the upper Snell envelope coincides with the minimal investment necessary for the purchase of a superhedging strategy at any time $t = 0, \dots, T$. Subsequently, we introduced the concept of quantile hedging. Thereby, for a fixed $\alpha \in (0, 1)$, we aimed to minimize the initial investment for a self-financing trading strategy which is a superhedging strategy for the claim H with probability α . This also helped us to understand the motivation behind the approach taken in the last part, namely by the construction of the neural network approximations. For the latter, we assumed that the filtration $\mathbb{F}$ is the natural filtration of a stochastic process, which allowed us to approximate predictable processes, and hence trading strategies, by multilayer feedforward neural networks, and to define the approximate superhedging and α -quantile hedging prices of H . Under the assumption that the superhedging price of H remains invariant when restricted to bounded strategies, we successfully approximated it by feedforward neural networks. Finally, we concluded the characterization of the superhedging price process by providing a neural network approximation of the process of consumption in the optional decomposition of the upper Snell envelope of H .

Due to the exceptional approximation properties of feedforward neural networks, the results presented in this thesis prove useful in cases where an explicit calculation of the superhedging and α -quantile hedging prices of a contingent claim is complicated.

Appendix A

Appendix

Proof of Theorem 2.21. (i) $\Rightarrow$ (ii): Suppose $Q \in \mathcal{P}$. Let $\bar{\xi} = (\xi^0, \xi)$ be a self-financing trading strategy with value process V , such that ξ is bounded. Then there exists a constant $c \in \mathbb{R}_+$ such that $|\xi^i| \leq c$, for $i = 1, \dots, d$. We recall that by Proposition 2.17,

$$V_t = V_0 + \sum_{k=1}^t \xi_k \cdot (X_k - X_{k-1}) \quad \mathbb{P}\text{-a.s., for } t = 0, \dots, T. \quad (\text{A.1})$$

Hence, for $t = 0, \dots, T$ we obtain

$$\begin{aligned} |V_t| &\leq |V_0| + \sum_{k=1}^t |\xi_k \cdot (X_k - X_{k-1})| \\ &\leq |V_0| + \sum_{k=1}^t c \left| \sum_{i=1}^d (X_k^i - X_{k-1}^i) \right| \\ &\leq |V_0| + \sum_{k=1}^t c \left(\sum_{i=1}^d (|X_k^i| + |X_{k-1}^i|) \right). \end{aligned} \quad (\text{A.2})$$

Since X is a Q -martingale, in particular we have $\mathbb{E}^Q[|X_k^i|] < \infty$, for $k = 0, \dots, T$ and $i = 1, \dots, d$, which combined with (A.2) yields

$$\mathbb{E}^Q[|V_t|] < \infty \quad \text{for } t = 0, \dots, T.$$

Moreover, using (A.1) and the fact that ξ is predictable and bounded, for $t = 1, \dots, T$ we obtain

$$\begin{aligned} \mathbb{E}^Q[V_t | \mathcal{F}_{t-1}] &= \mathbb{E}^Q[V_{t-1} + \xi_t \cdot (X_t - X_{t-1}) | \mathcal{F}_{t-1}] \\ &= V_{t-1} + \xi_t \cdot \mathbb{E}^Q[X_t - X_{t-1} | \mathcal{F}_{t-1}] \\ &= V_{t-1} + \sum_{i=1}^d \xi_t^i \mathbb{E}^Q[X_t^i - X_{t-1}^i | \mathcal{F}_{t-1}] \\ &= V_{t-1} \quad \mathbb{P}\text{-a.s.,} \end{aligned}$$

which proves that V is a Q -martingale.

(ii) $\Rightarrow$ (iii): Suppose condition (ii) holds. Let $\bar{\xi} = (\xi^0, \xi)$ be a self-financing

trading strategy with value process V . We prove the following implication.

$$\text{For any } t = 1, \dots, T : \text{ if } \mathbb{E}^Q[V_t^-] < \infty, \text{ then } \mathbb{E}^Q[V_t | \mathcal{F}_{t-1}] = V_{t-1}. \quad (\text{A.3})$$

We note that since $\mathbb{E}^Q[V_t^-] < \infty$, $\mathbb{E}^Q[V_t | \mathcal{F}_{t-1}]$ is well-defined on $(-\infty, \infty]$. For $a > 0$, let $\xi_t^{(a)} := \xi_t \mathbb{1}_{\{|\xi_t| \leq a\}}$. Then by Remark 2.18, $(V_0, \xi^{(a)})$ defines a self-financing trading strategy with value process $V^{(a)}$ given by

$$V_t^{(a)} = V_0 + \sum_{k=1}^t \xi_k^{(a)} \cdot (X_k - X_{k-1}) \quad \mathbb{P}\text{-a.s., for } t = 0, \dots, T. \quad (\text{A.4})$$

Moreover, since $\xi^{(a)}$ is bounded, $V^{(a)}$ is a Q -martingale by (ii). Therefore we get

$$\begin{aligned} \mathbb{1}_{\{|\xi_t| \leq a\}} \mathbb{E}^Q[V_t | \mathcal{F}_{t-1}] &= \mathbb{E}^Q[\mathbb{1}_{\{|\xi_t| \leq a\}} V_t | \mathcal{F}_{t-1}] - \mathbb{E}^Q[V_t^{(a)} - V_{t-1}^{(a)} | \mathcal{F}_{t-1}] \\ &= \mathbb{E}^Q[\mathbb{1}_{\{|\xi_t| \leq a\}} V_t - \xi_t^{(a)} \cdot (X_t - X_{t-1}) | \mathcal{F}_{t-1}] \\ &= \mathbb{E}^Q[\mathbb{1}_{\{|\xi_t| \leq a\}} (V_t - \xi_t \cdot (X_t - X_{t-1})) | \mathcal{F}_{t-1}] \\ &= \mathbb{E}^Q[\mathbb{1}_{\{|\xi_t| \leq a\}} V_{t-1} | \mathcal{F}_{t-1}] \\ &= \mathbb{1}_{\{|\xi_t| \leq a\}} V_{t-1} \quad \mathbb{P}\text{-a.s.,} \end{aligned}$$

where we used (A.4) in the second equality, and (A.1) in the penultimate equality. Letting $a \rightarrow \infty$ proves (A.3). Suppose now

$$\mathbb{E}^Q[V_T^-] < \infty.$$

Then, by (A.3) and Jensen's inequality for conditional expectations, we have

$$\mathbb{E}^Q[V_{T-1}^-] = \mathbb{E}^Q[\mathbb{E}^Q[V_T | \mathcal{F}_{T-1}]^-] \leq \mathbb{E}^Q[\mathbb{E}^Q[V_T^- | \mathcal{F}_{T-1}]] = \mathbb{E}^Q[V_T^-] < \infty.$$

By backward induction it follows that $\mathbb{E}^Q[V_t] < \infty$, for $t = 0, \dots, T$, which combined with (A.3) implies

$$\mathbb{E}^Q[V_t | \mathcal{F}_{t-1}] = V_{t-1}, \quad \text{for } t = 1, \dots, T.$$

In particular, $\mathbb{E}^Q[V_t] = V_0$ is a finite constant, so $\mathbb{E}^Q[V_t^-] < \infty$ implies $\mathbb{E}^Q[|V_t|] < \infty$ for all t . Hence we conclude that V is a Q -martingale.

(iii) $\Rightarrow$ (iv): Suppose (iii) holds. Let $\bar{\xi} = (\xi^0, \xi)$ be a self-financing trading strategy with value process V such that $V_T \geq 0$ Q -a.s. Then clearly $\mathbb{E}^Q[V_T] < \infty$, so V is a Q -martingale by (iii). Hence it follows that

$$\mathbb{E}^Q[V_T] = \mathbb{E}^Q[V_T | \mathcal{F}_0] = V_0.$$

(iv) $\Rightarrow$ (i): Suppose (iv) holds, and fix $i \in \{1, \dots, d\}$ and $t \in \{1, \dots, T\}$. We prove Q -integrability of X_t^i . For $j = 1, \dots, d$ we define the d -dimensional process ξ by

$$\xi_s^j := \begin{cases} \mathbb{1}_{\{s \leq t\}} & \text{if } j = i, \\ 0 & \text{if } j \neq i, \end{cases}$$

for $s = 0, \dots, T$. Then ξ is predictable, since $\mathbb{1}_{\{s \leq t\}} = \mathbb{1}_{\{s-1 < t\}}$. Hence, (X_0^i, ξ) describes a self-financing trading strategy with value process V . By definition of ξ , we get

$$\begin{aligned} V_T &= V_0 + \sum_{s=1}^T \xi_s \cdot (X_s - X_{s-1}) = V_0 + \sum_{s=1}^T \xi_s^i (X_s^i - X_{s-1}^i) \\ &= V_0 + X_T^i - X_0^i = X_T^i \geq 0 \quad Q\text{-a.s.} \end{aligned}$$

Then (iv) yields

$$\mathbb{E}^Q[X_t^i] = \mathbb{E}^Q[V_T] = V_0 = X_0^i, \quad (\text{A.5})$$

and therefore $X_t^i \in \mathcal{L}^1(Q)$. It remains to show that $\mathbb{E}^Q[X_{t-1}^i | \mathcal{F}_{t-1}] = X_t^i$. Let $A \in \mathcal{F}_{t-1}$. We define a d -dimensional predictable process η by

$$\eta_s^j := \begin{cases} \mathbb{1}_{\{s < t\}} + \mathbb{1}_{A^C} \mathbb{1}_{\{s=t\}} & \text{if } j = i, \\ 0 & \text{if } j \neq i, \end{cases}$$

for $s = 1, \dots, T$. As above, we consider the self-financing trading strategy $\tilde{\eta}$ defined by η and the initial investment $\tilde{V}_0 := X_0^i$, and denote its value process by $(\tilde{V}_s)_{s=0, \dots, T}$. Then we get

$$\begin{aligned} \tilde{V}_T &= \tilde{V}_0 + \sum_{s=1}^T \eta_s \cdot (X_s - X_{s-1}) = \tilde{V}_0 + \sum_{s=1}^T \eta_s^i (X_s^i - X_{s-1}^i) \\ &= \tilde{V}_0 + X_{t-1}^i - X_0^i + \mathbb{1}_{A^C} (X_t^i - X_{t-1}^i) = \mathbb{1}_{A^C} X_t^i + \mathbb{1}_A X_{t-1}^i \geq 0 \quad Q\text{-a.s.} \end{aligned}$$

Using (iv), we obtain

$$X_0^i = \tilde{V}_0 = \mathbb{E}^Q[\tilde{V}_T] = \mathbb{E}^Q[\mathbb{1}_{A^C} X_t^i] + \mathbb{E}^Q[\mathbb{1}_A X_{t-1}^i]. \quad (\text{A.6})$$

Finally, combining (A.5) and (A.6) yields

$$\begin{aligned} \mathbb{E}^Q[\mathbb{1}_A X_t^i] &= \mathbb{E}^Q[X_t^i] - \mathbb{E}^Q[\mathbb{1}_{A^C} X_t^i] \\ &= X_0^i - (X_0^i - \mathbb{E}^Q[\mathbb{1}_A X_{t-1}^i]) \\ &= \mathbb{E}^Q[\mathbb{1}_A X_{t-1}^i], \end{aligned}$$

and since $A \in \mathcal{F}_{t-1}$ was arbitrary this shows $\mathbb{E}^Q[X_{t-1}^i | \mathcal{F}_{t-1}] = X_t^i$. As i and t were arbitrary, we conclude that X is a Q -martingale. $\square$

Proof of Proposition 4.9. The proof is similar to the proof of Theorem 4.4. Let $(\alpha_n)_{n \in \mathbb{N}} \subset (0, 1)$ be an increasing sequence with $\lim_{n \rightarrow \infty} \alpha_n = 1$. Then $\mathcal{V}^{\alpha_{n+1}} \subseteq \mathcal{V}^{\alpha_n}$, so since $\varphi_V \leq 1$ we have

$$\inf_{V \in \mathcal{V}^{\alpha_n}} \sup_{\mathbb{P}^* \in \mathcal{D}} \mathbb{E}^*[H \varphi_V] \leq \inf_{V \in \mathcal{V}^{\alpha_{n+1}}} \sup_{\mathbb{P}^* \in \mathcal{D}} \mathbb{E}^*[H \varphi_V] \leq \sup_{\mathbb{P}^* \in \mathcal{D}} \mathbb{E}^*[H] < \infty.$$

Thus, the sequence $(\inf_{V \in \mathcal{V}^{\alpha_n}} \sup_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[H\varphi_V])_{n \in \mathbb{N}}$ is increasing and bounded, so its limit exists and satisfies

$$\lim_{n \rightarrow \infty} \inf_{V \in \mathcal{V}^{\alpha_n}} \sup_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[H\varphi_V] \leq \sup_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[H]. \quad (\text{A.7})$$

We show the converse inequality in (A.7). By Proposition 4.8, for each $n \in \mathbb{N}$ there exists an admissible strategy with value process $V_n \in \mathcal{V}^{\alpha_n}$ such that

$$\mathbb{E}[\varphi_{V_n}] = \alpha_n \quad \text{and} \quad \inf_{V \in \mathcal{V}^{\alpha_n}} \sup_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[H\varphi_V] = \sup_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[H\varphi_{V_n}]. \quad (\text{A.8})$$

Hence, it suffices to show

$$\sup_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[H] \leq \lim_{n \rightarrow \infty} \sup_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[H\varphi_{V_n}].$$

Since $(\varphi_{V_n})_{n \in \mathbb{N}} \subset L^\infty(\Omega, \mathcal{F}_T, \mathbb{P}; [0, 1])$, by Lemma 2.8 there exists a sequence $(\psi_n)_{n \in \mathbb{N}}$, given by

$$\psi_n = \sum_{k=n}^{\infty} \lambda_k^n \varphi_{V_k} \in \text{conv}(\varphi_{V_n}, \varphi_{V_{n+1}}, \dots) \quad (\text{A.9})$$

for some $(\lambda_k^n)_{k=n}^{\infty} \geq 0$ with $\sum_{k=n}^{\infty} \lambda_k^n = 1$ and only finitely many $\lambda_k^n \neq 0$ for each $n \in \mathbb{N}$, and such that (A.9) converges $\mathbb{P}$ -a.s. to some $\psi \in L^\infty(\Omega, \mathcal{F}_T, \mathbb{P}; [0, 1])$. Then using dominated convergence and (A.8) we obtain

$$\begin{aligned} \mathbb{E}[\psi] &= \lim_{n \rightarrow \infty} \mathbb{E}[\psi_n] = \lim_{n \rightarrow \infty} \mathbb{E}\left[\sum_{k=n}^{\infty} \lambda_k^n \varphi_{V_k}\right] = \lim_{n \rightarrow \infty} \left(\sum_{k=n}^{\infty} \lambda_k^n \mathbb{E}[\varphi_{V_k}]\right) \\ &\geq \lim_{n \rightarrow \infty} \left(\sum_{k=n}^{\infty} \lambda_k^n \inf_{l \geq n} \mathbb{E}[\varphi_{V_l}]\right) = \lim_{n \rightarrow \infty} \inf \mathbb{E}[\varphi_{V_n}] = \lim_{n \rightarrow \infty} \alpha_n = 1. \end{aligned}$$

Since $0 \leq \psi \leq 1$, it follows that $\psi = 1$ $\mathbb{P}$ -a.s. Now fix an arbitrary $\tilde{\mathbb{P}} \in \mathcal{P}$. Again by dominated convergence, we have

$$\begin{aligned} \tilde{\mathbb{E}}[H] &= \tilde{\mathbb{E}}[H\psi] = \lim_{n \rightarrow \infty} \tilde{\mathbb{E}}[H\psi_n] = \lim_{n \rightarrow \infty} \tilde{\mathbb{E}}\left[H \sum_{k=n}^{\infty} \lambda_k^n \varphi_{V_k}\right] \\ &= \lim_{n \rightarrow \infty} \left(\sum_{k=n}^{\infty} \lambda_k^n \tilde{\mathbb{E}}[H\varphi_{V_k}]\right) \leq \lim_{n \rightarrow \infty} \left(\sum_{k=n}^{\infty} \lambda_k^n \sup_{l \geq n} \tilde{\mathbb{E}}[H\varphi_{V_l}]\right) \\ &= \limsup_{n \rightarrow \infty} \tilde{\mathbb{E}}[H\varphi_{V_n}] \leq \lim_{n \rightarrow \infty} \sup_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[H\varphi_{V_n}]. \end{aligned}$$

Since $\tilde{\mathbb{P}} \in \mathcal{P}$ was arbitrary, it follows that

$$\sup_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[H] \leq \lim_{n \rightarrow \infty} \sup_{\mathbb{P}^* \in \mathcal{P}} \mathbb{E}^*[H\varphi_{V_n}],$$

which together with (A.7) shows

$$\lim_{n \rightarrow \infty} \inf_{V \in \mathcal{V}^{\alpha n}} \sup_{\mathbb{P}^* \in \mathcal{D}} \mathbb{E}^*[H\varphi_V] = \sup_{\mathbb{P}^* \in \mathcal{D}} \mathbb{E}^*[H]$$

and thus concludes the proof. $\square$

Proof of Corollary 5.3. Fix $\delta, \varepsilon > 0$ and choose $C > 0$ such that

$$\mu[\{x \in \mathbb{R}^d : \|f(x)\| > C\}] < \frac{\varepsilon}{2}. \quad (\text{A.10})$$

Let $A := \{x \in \mathbb{R}^d : \|f(x)\| \leq C\}$. For $\tilde{f} := \mathbb{1}_A f$ we have $\tilde{f} \in L^1(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d), \mu)$, and hence by Theorem 5.2 (extended for the multioutput case), there exists a neural network $g \in \mathcal{NN}_{d,m}^\sigma$ such that

$$\|\tilde{f} - g\|_{L^1(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d), \mu)} = \int_{\mathbb{R}^d} \|\tilde{f}(x) - g(x)\| \mu(dx) < \frac{\varepsilon \delta}{4}.$$

By Markov's inequality, this implies

$$\mu\left[\left\{x \in \mathbb{R}^d : \|\tilde{f}(x) - g(x)\| > \frac{\delta}{2}\right\}\right] \leq \frac{2}{\delta} \int_{\mathbb{R}^d} \|\tilde{f}(x) - g(x)\| \mu(dx) < \frac{\varepsilon}{2}. \quad (\text{A.11})$$

Since $f - \tilde{f} = f \mathbb{1}_{A^c}$, the triangle inequality combined with (A.10) and (A.11) yield

$$\begin{aligned} & \mu[\{x \in \mathbb{R}^d : \|f(x) - g(x)\| > \delta\}] \\ & \leq \mu[\{x \in \mathbb{R}^d : \|f(x) - \tilde{f}(x)\| + \|\tilde{f}(x) - g(x)\| > \delta\}] \\ & \leq \mu\left[\left\{x \in \mathbb{R}^d : \|f(x) - \tilde{f}(x)\| > \frac{\delta}{2}\right\} \cup \left\{x \in \mathbb{R}^d : \|\tilde{f}(x) - g(x)\| > \frac{\delta}{2}\right\}\right] \\ & < \mu[\{x \in \mathbb{R}^d : \|f(x)\| > C\}] + \frac{\varepsilon}{2} \\ & < \varepsilon, \end{aligned}$$

which concludes the proof. $\square$

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